LFSUPC PHYSICS WORKING GROUP • MONDAY, AUGUST 10, 2026

π/K/p Spectra Analysis

$\sqrt{s_{NN}}$=200 GeV O+O Collisions for Space Radiation Studies

Andrew Liggett

AGENDA

What We'll Cover

1. **Physics motivation for O+O**
   Oxygen as a galactic cosmic ray proxy -- why O+O collisions matter for space radiation and shielding models.

2. **Detector, tracking, and event-quality checks**
   Acceptance, PID, dE/dx and BTOF fit diagnostics, and event QA.

3. **Yield extraction and corrected spectra**
   From raw TPC/BTOF yields through corrected mT-m0 spectra -- now complete for π/K/p.

4. **Status and next steps**
   Where the analysis stands -- eTOF calibration ongoing, embedding request next.

Space Radiation Motivation

1. Oxygen as a galactic cosmic ray proxy

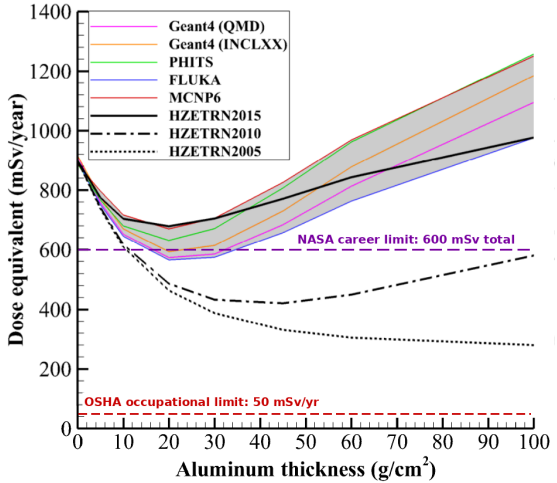
Oxygen is a major heavy-ion component of galactic cosmic rays -- O+O collisions probe the same nuclear fragmentation GCRs undergo in spacecraft shielding and human tissue.

2. Benchmarking radiation transport models

Identified π/K/p yields provide direct data to benchmark hadronic interaction models (FLUKA, GEANT4, PHITS) used to predict astronaut radiation dose.

3. Informing shielding and mission risk models

Better-constrained particle-production yields feed into spacecraft shielding design and astronaut dose/risk estimates for lunar and Mars exploration.



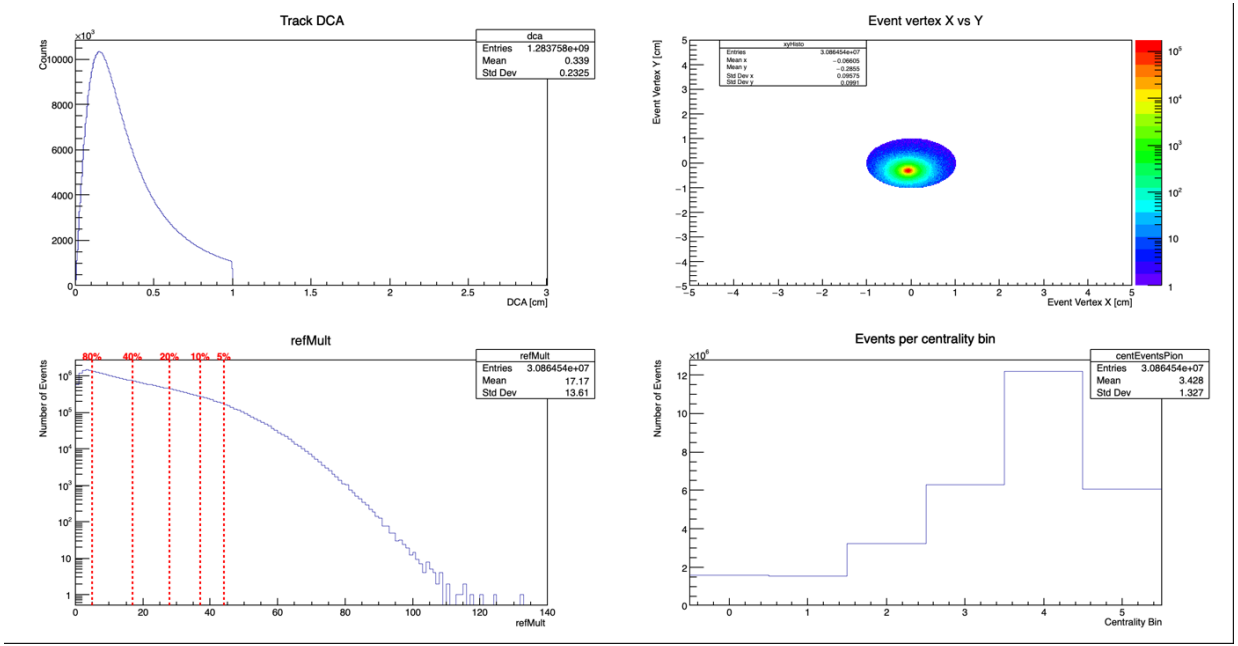
(a)

Dose equivalent vs. aluminum shielding -- Monte Carlo codes (incl. FLUKA) vs. HZETRN versions. Norbury, Slaba et al. (2019); OSHA and NASA limit lines added (NASA's is a career total, not annual).

Event Selection Cuts

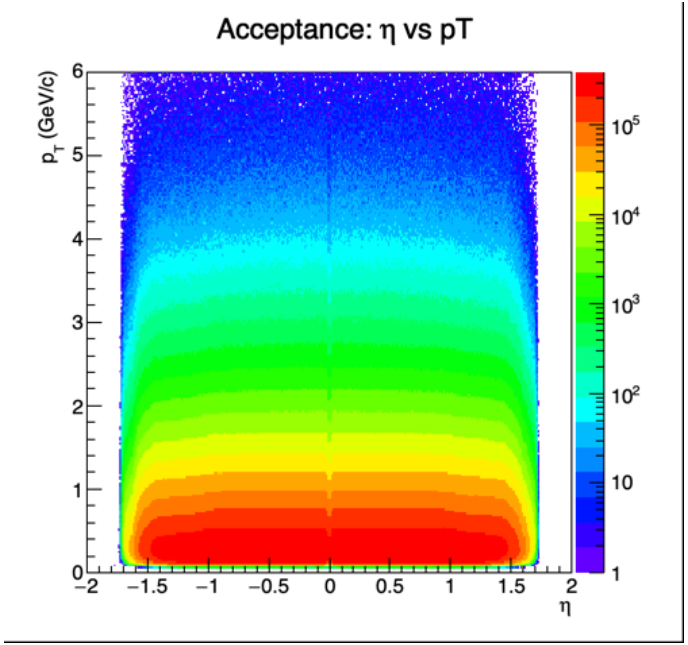
Cut	Value
Vertex Vz	[-2, 2] cm
Vertex Vr	<= 1.0 cm
Trigger ID	860003
Global DCA	<= 1.0 cm
Centrality Bins	0-5%, 5-10%, 10-20%, 20-40%, 40-80%, 80-100% (raw refMult)

Event QA: Trigger Cut Verified



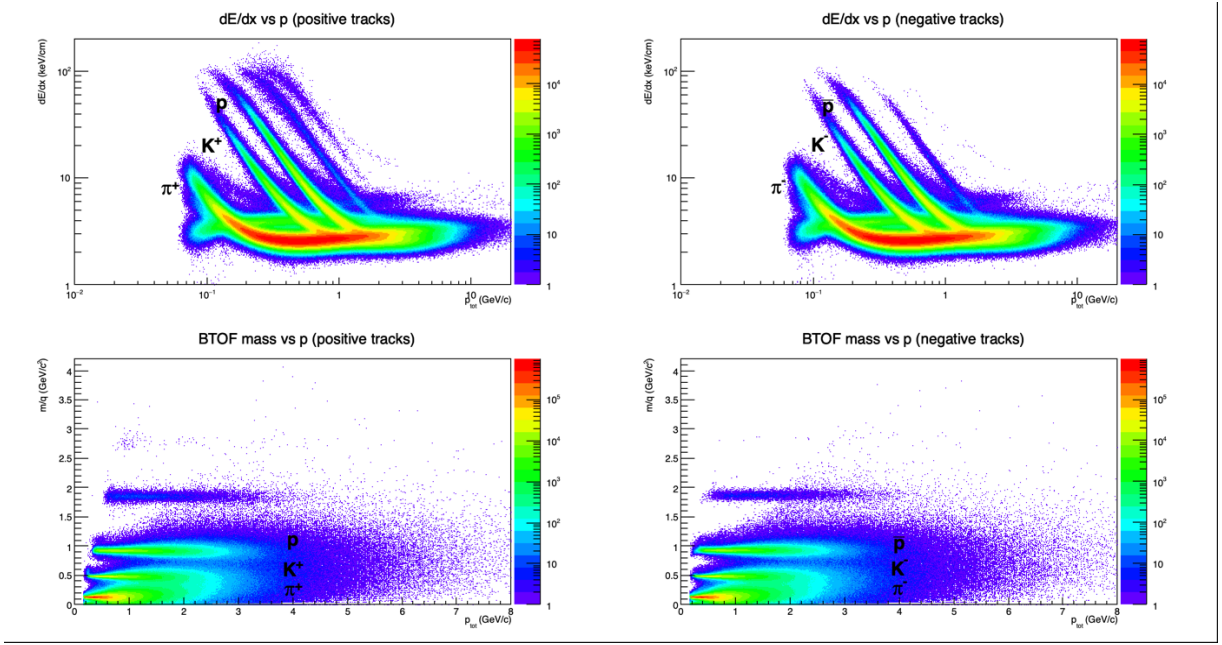
PION event QA: DCA, vertex XY, refMult centrality cuts, and events per centrality bin.

Acceptance: η vs pT



TPC-only acceptance in pseudorapidity (eta) vs pT -- defines the kinematic reach of the analysis.

Particle ID: dE/dx and BTOF Mass

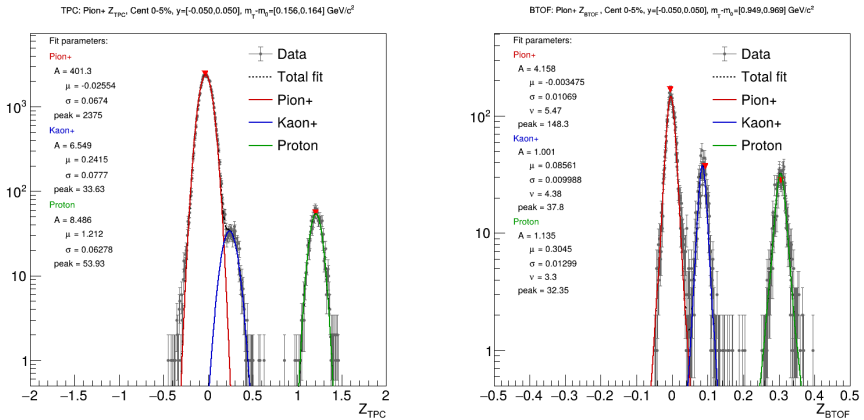


TPC dE/dx vs p and BTOF mass vs p for positive and negative tracks -- pi/K/p bands are cleanly separated.

Z-Fitter Example: dE/dx Fit Diagnostics

$$Z_{\mathrm{TPC}} = \ln\left[\frac{(dE/dx)_{\mathrm{meas}}}{(dE/dx)_{\mathrm{Bichsel}}(p,\ \mathrm{species})}\right]$$

$$Z_{\mathrm{BTOF}} = \left(\frac{1}{\beta}\right)_{\mathrm{meas}} - \left(\frac{1}{\beta}\right)_{\mathrm{expected}}(p,\ \mathrm{mass}) \qquad \text{where } \left(\frac{1}{\beta}\right)_{\mathrm{expected}}(p, m) = \sqrt{1 + \frac{m^2}{p^2}}$$

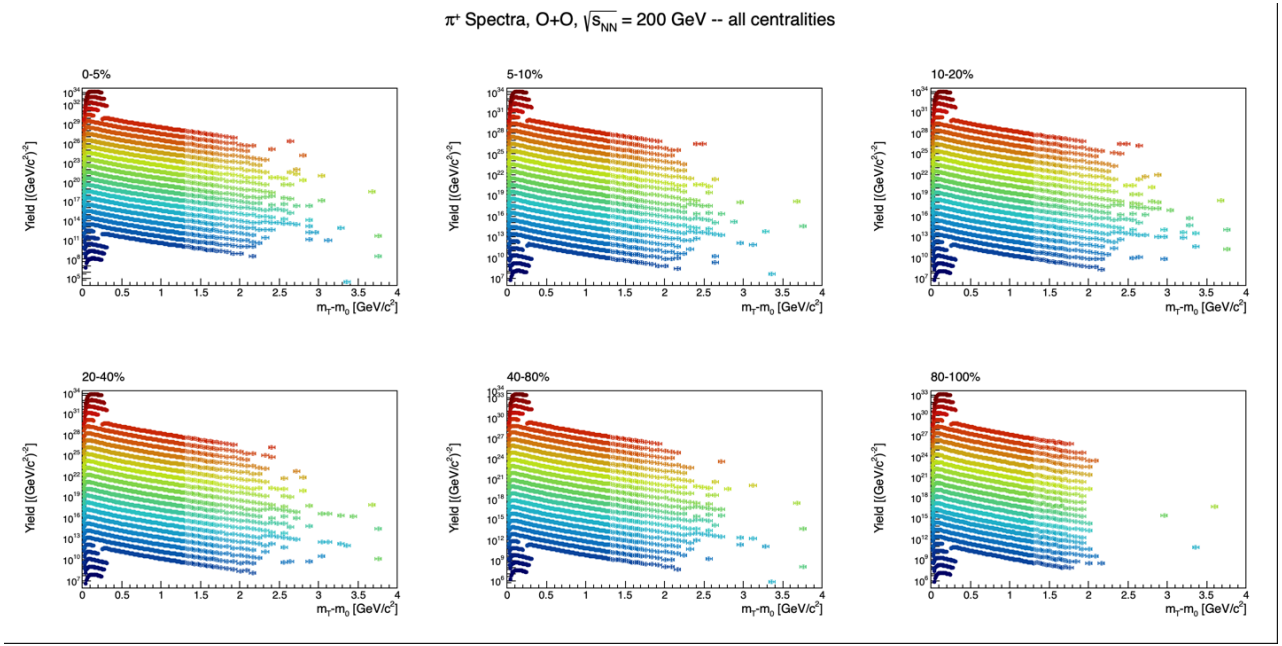


Example Gaussian (TPC) and Student's-t (BTOF) fits to pion-yield data -- pion+/kaon+/proton components, 0-5% centrality.

Spectra Corrections: Status

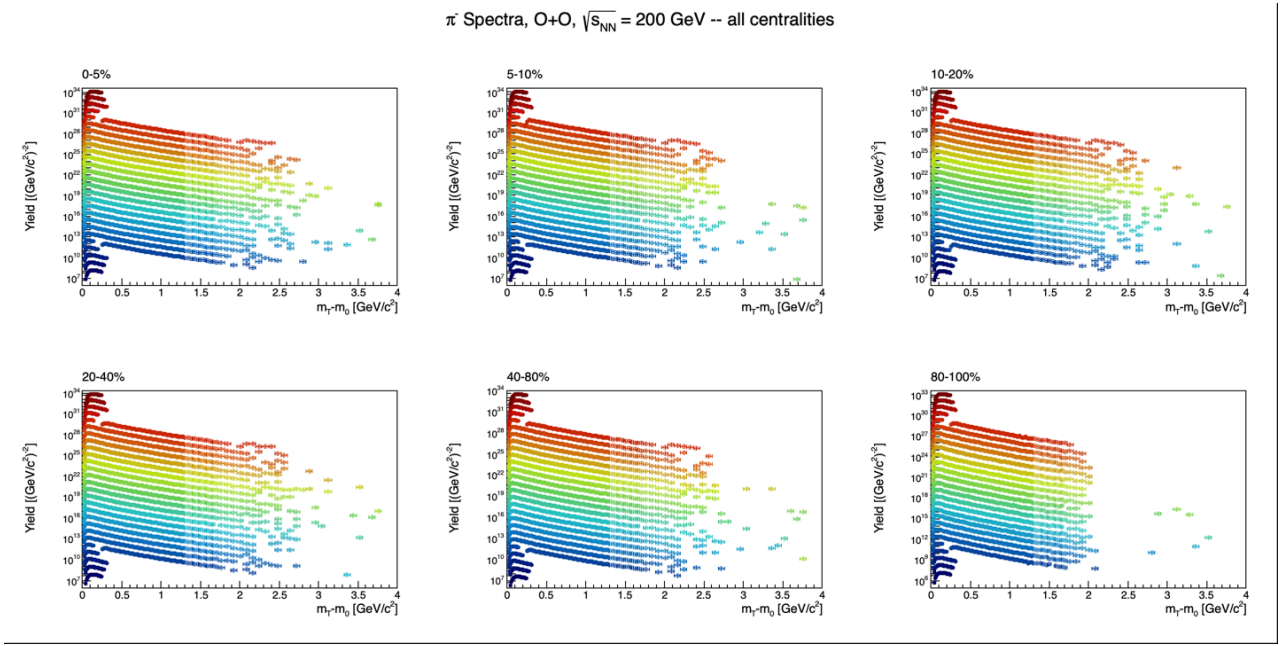
Correction	Status	Notes
BTOF efficiency	**Done**	Data-driven: ratio of TPC-identified tracks with a matching BTOF hit to all TPC-identified tracks, fit vs mT-m0. No embedding needed.
TPC efficiency	Not yet applied	Needs an embedding-derived efficiency file -- still missing.
Energy loss	Not yet applied	Same embedding-derived file as TPC efficiency -- applied in the same call, still missing.
Hybrid feed-down	Not yet applied	Also needs the same embedding-derived file; will be enabled once it exists.

Pion+: Combined All-Centralities Spectra View



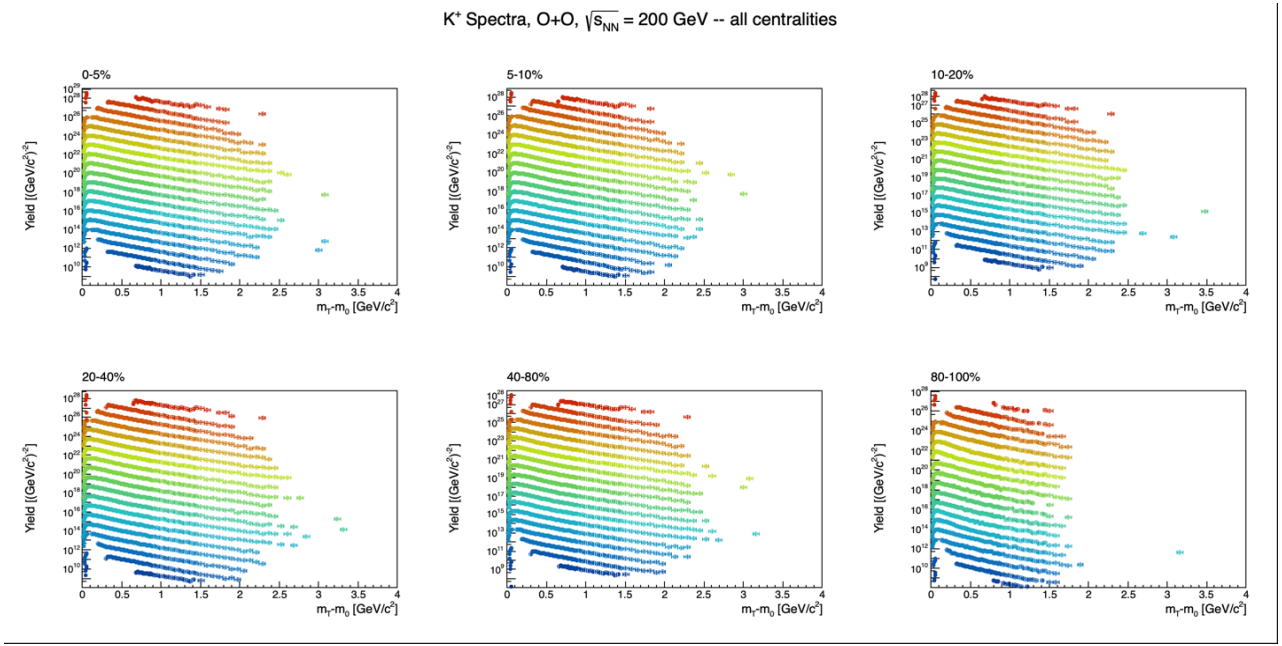
BTOF-corrected pion+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Pion-: Combined All-Centralities Spectra View



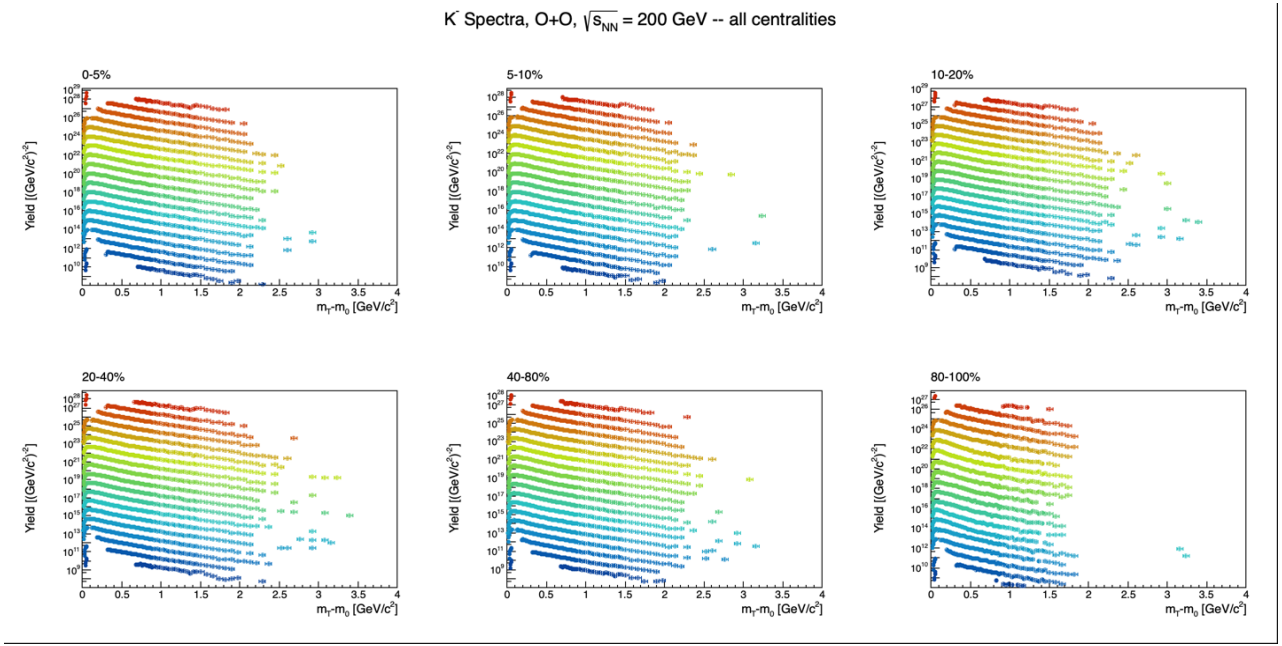
BTOF-corrected pion- mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Kaon+: Combined All-Centralities Spectra View



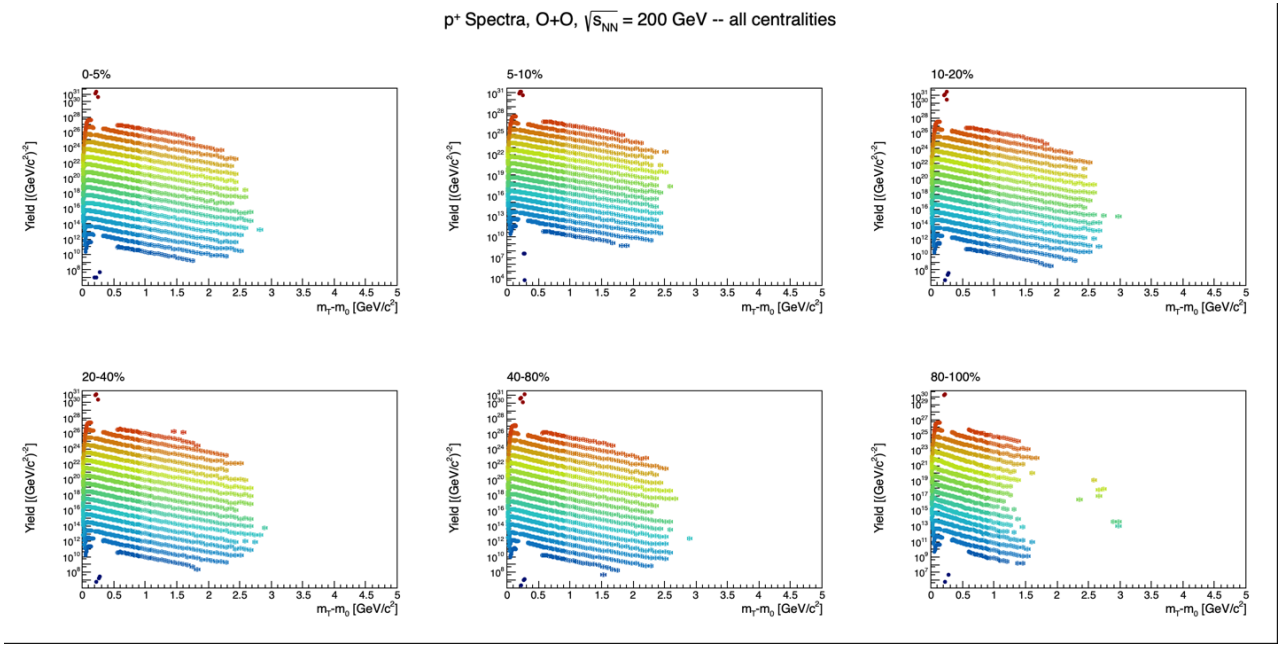
BTOF-corrected kaon+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Kaon-: Combined All-Centralities Spectra View



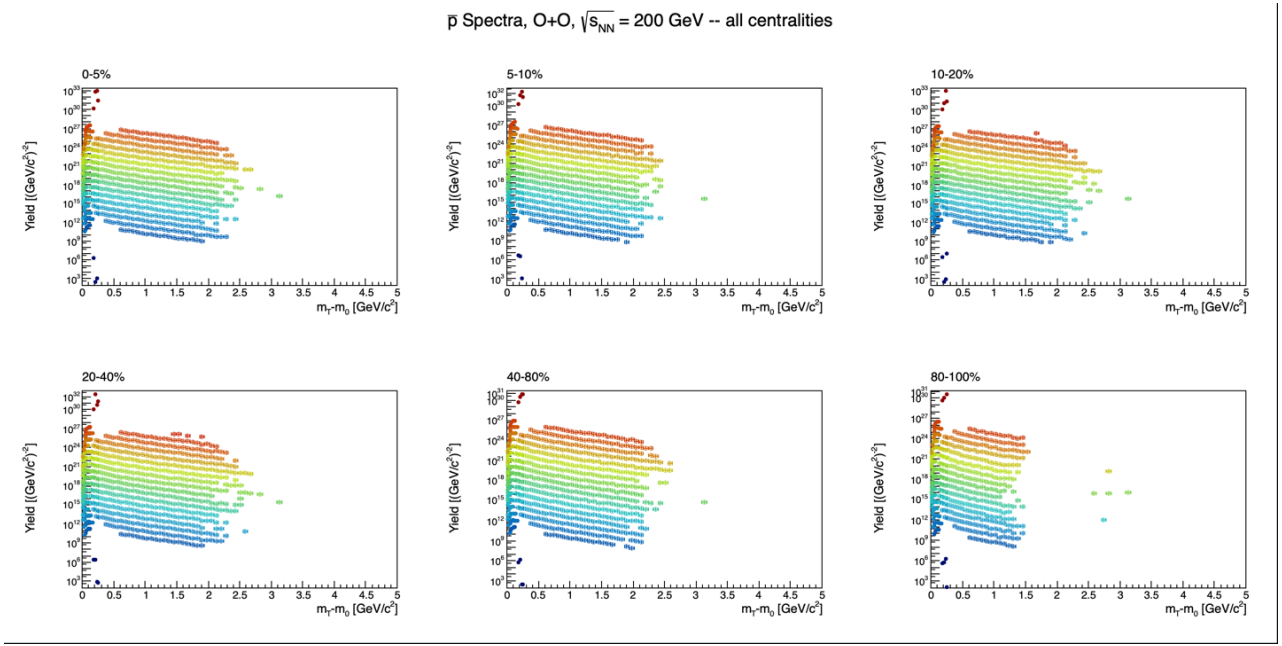
BTOF-corrected kaon- mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Proton+: Combined All-Centralities Spectra View



BTOF-corrected proton+ mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Antiproton: All-Centralities Spectra View



BTOF-corrected antiproton mT-m0 spectra, TPC+BTOF combined, all six centrality bins, all rapidity bins together.

Embedding Request Checklist

A) Real Data Sample

Item	What We Know
Trigger set name	production_OO_200GeV_2021
File type	st_*_adc DAQ (fixed)
Production tag	P24iy
Run range	TBD
Trigger IDs	860003
Vertex cuts	|Vz|<2cm, Vr<1cm
Event cuts	refMult, 6 cent bins

B) Simulation & Reconstruction

Item	What We Know
Particle & decay	π±/K±/p (stable)
pT range/dist.	up to ~4 GeV/c
η or y range	|y| < 1.3 (populated)
MC particles/event	TBD
Event generator	N/A (single-particle)
Recon. requirements	SL24y, ColliderCenter
EMC/BTOF sim.	BTOF not needed (data-driven)

Criteria: STAR embedding tutorial -- star-juniors.github.io/tutorials/embedding.html#how-to-create-an-embedding-request-what-you-must-prepare. Submitted via PWG convener once prepared.

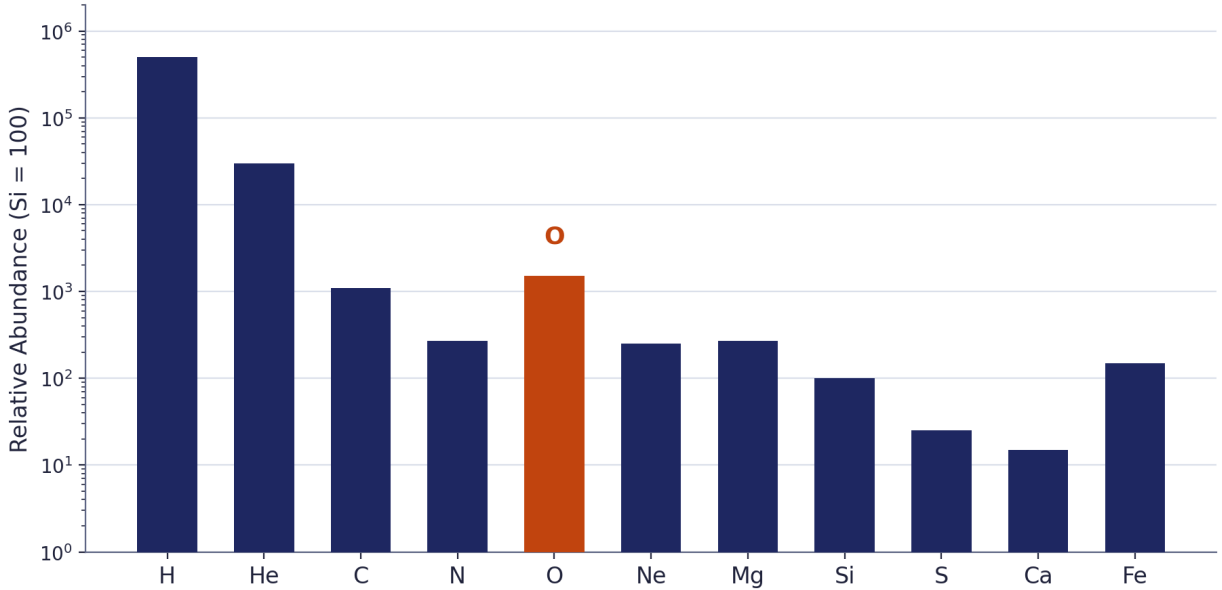
Summary & Next Steps

- **Motivation** -- O+O collisions as a galactic cosmic ray proxy for space radiation and shielding studies.
- **BTOF-corrected spectra produced for π/K/p** -- all six centrality bins, TPC+BTOF combined.
- **Currently working on eTOF calibrations** -- MuDsts restored, calibration pass underway.
- **Submit an embedding request** -- remaining corrections on hold until the embedding is produced for this production.





GCR Elemental Composition

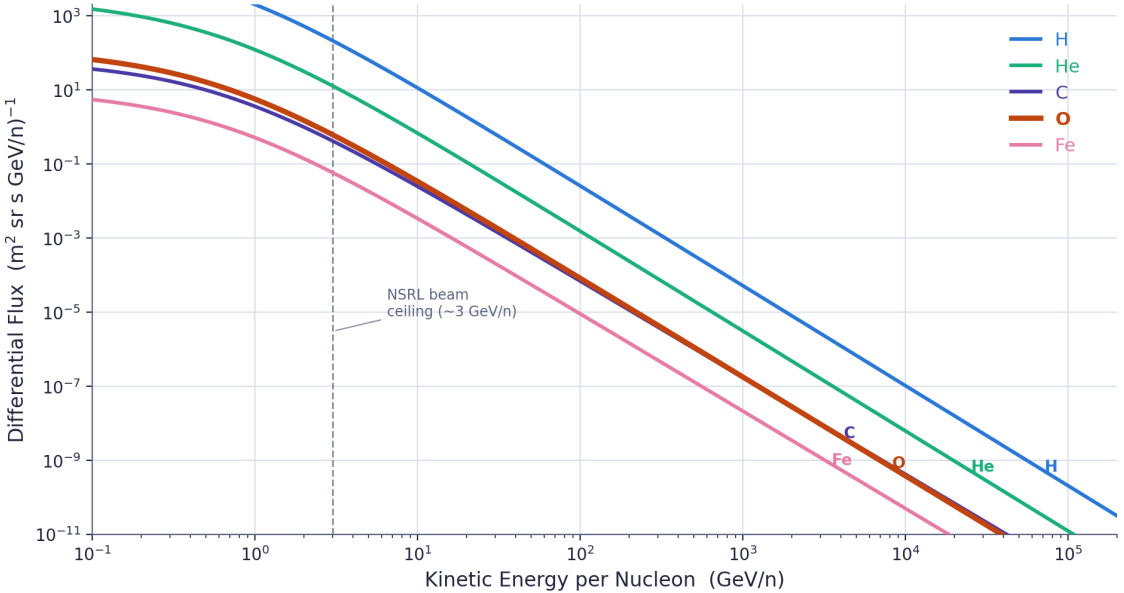


Oxygen is a major primary GCR component and spallation parent -- directly relevant to the O+O system studied here.

Source: Simpson (1983), Ann. Rev. Nucl. Part. Sci. 33:323; ACE-CRIS (Caltech ACE News #83). Representative reconstruction, not a digitized figure.

BACKUP

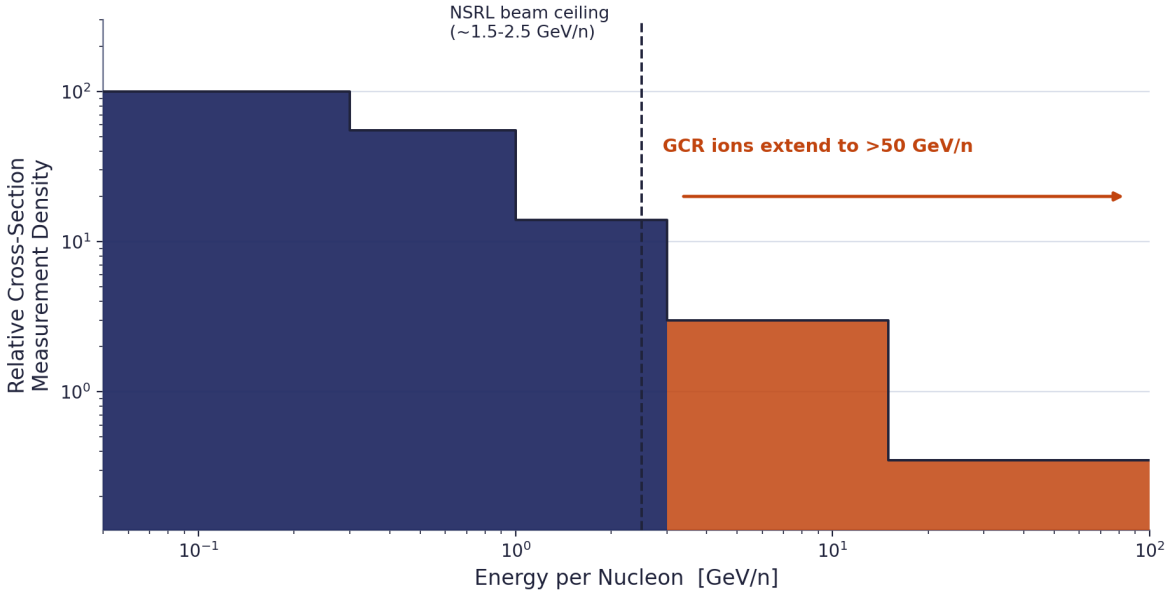
GCR Energy Spectrum by Element



Sources: PDG "Cosmic Rays" review; Ahn et al. 2009, ApJ 707:593 (CREAM); Simpson (1983) abundances. Representative reconstruction, not a digitized figure.

BACKUP

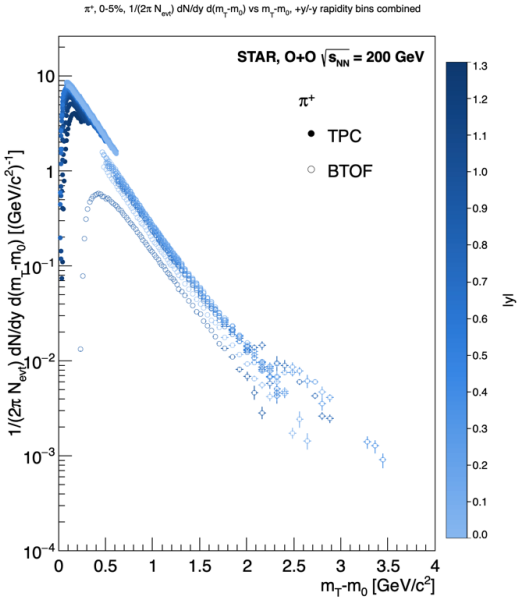
Nuclear Data Coverage Gap



FLUKA/GEANT4/PHITS are tuned to measured cross sections -- data thins sharply above ~3 GeV/n, beyond NSRL's beam ceiling.

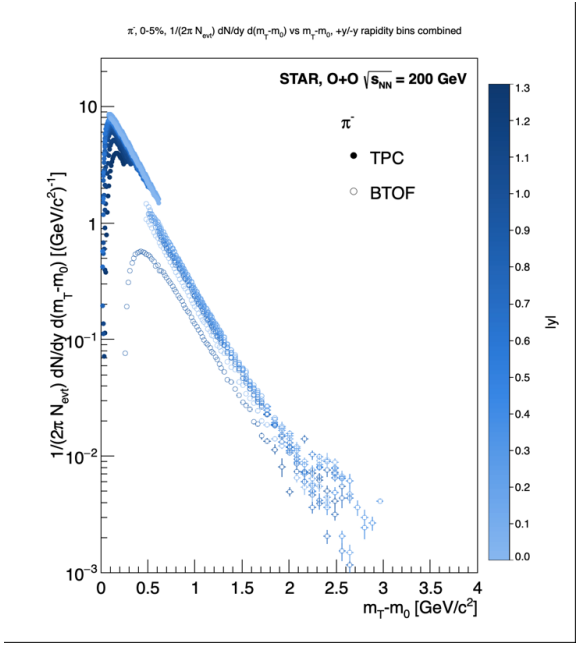
Source: Norbury & Losekamm, NASA NTRS 20240015733; NSRL beam specs, BNL. Representative reconstruction, not a digitized figure.

Raw Yield Extraction: Z-Fitter Output (π+)



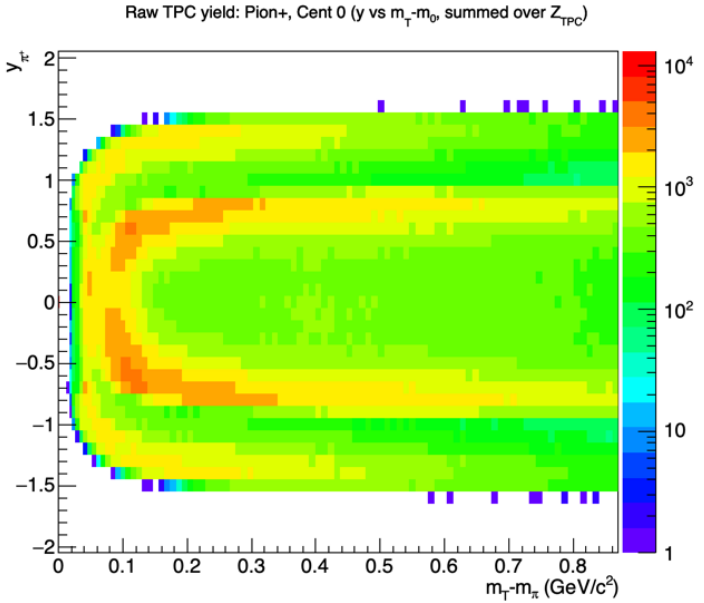
TPC vs BTOF raw dN/dy vs mT-m0 for pion+, 0-5% centrality -- before the two detectors are merged and corrected.

Raw Yield Extraction: Z-Fitter Output (π−)



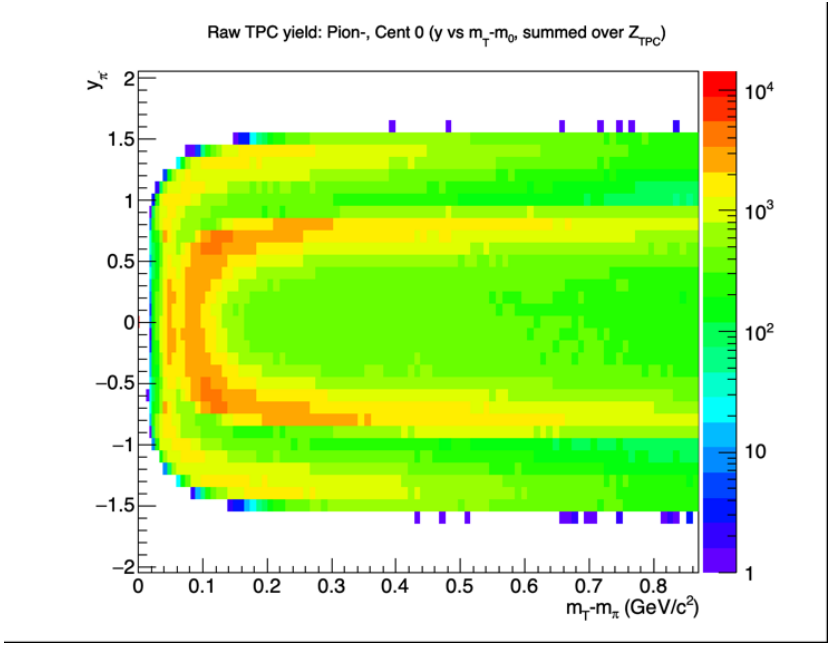
TPC vs BTOF raw dN/dy vs mT-m0 for pion-, 0-5% centrality -- before the two detectors are merged and corrected.

Raw TPC Yield: Rapidity vs mT-m0 (π+)



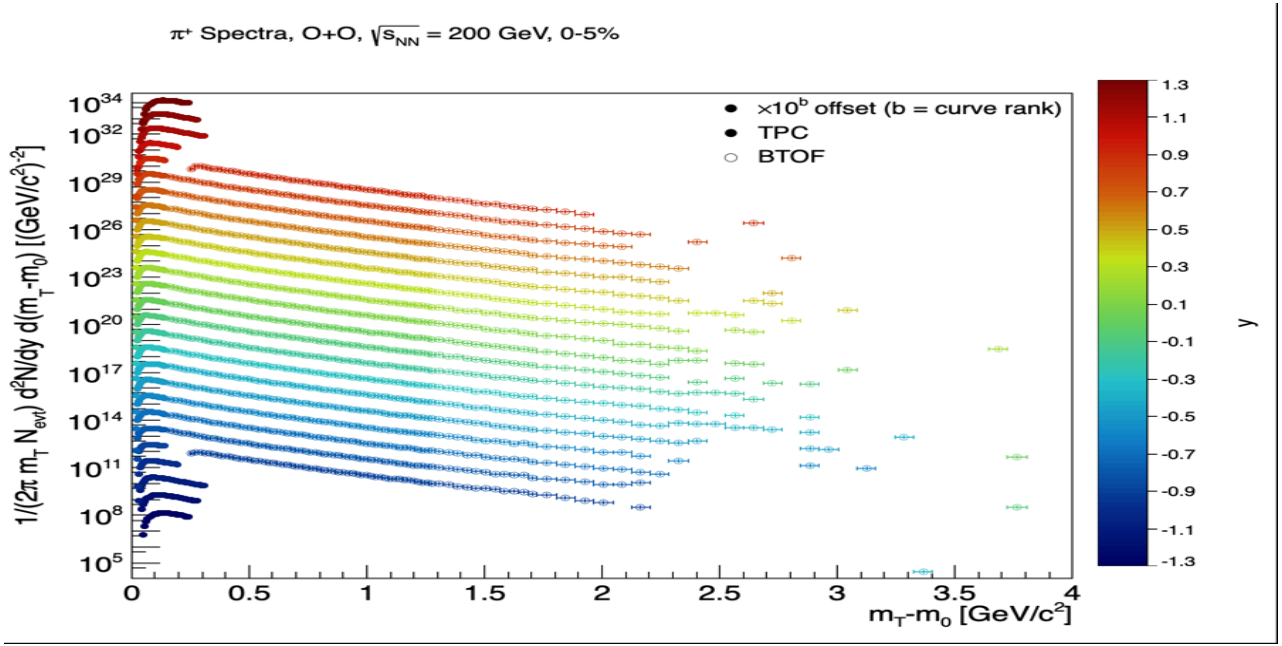
Pion+ raw TPC yield vs rapidity and mT-m0, 0-5% centrality, summed over Z_TPC -- input to the corrections.

Raw TPC Yield: Rapidity vs mT-m0 (π−)



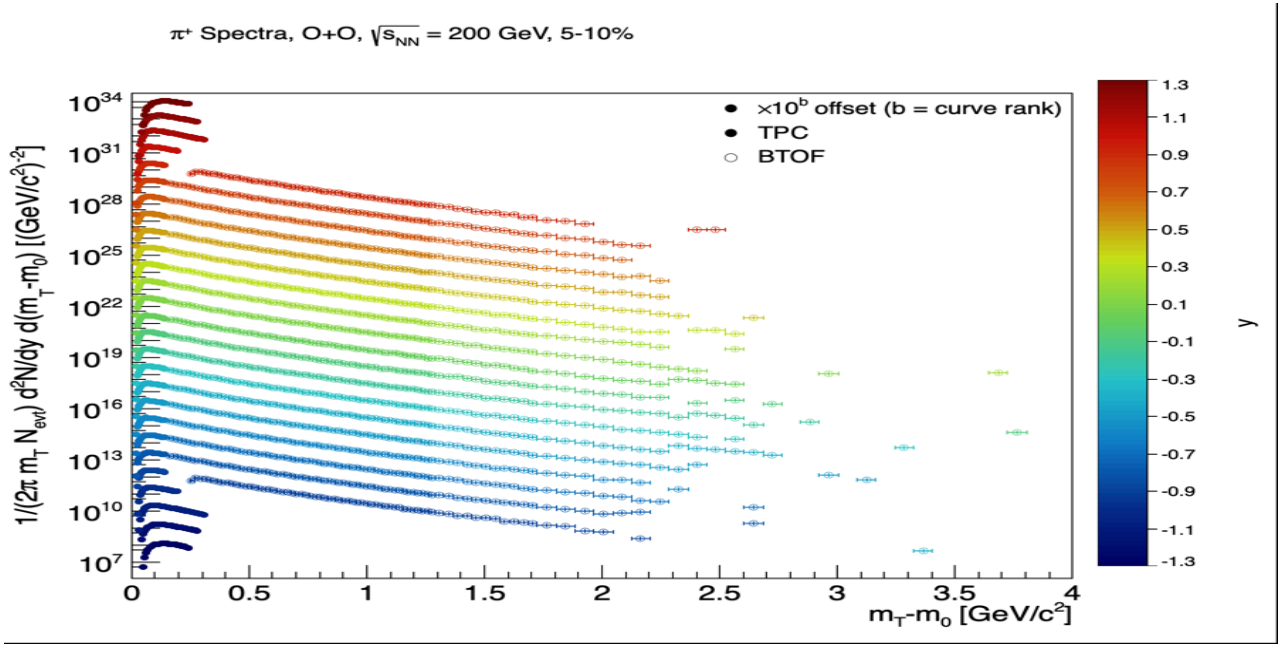
Pion- raw TPC yield vs rapidity and mT-m0, 0-5% centrality, summed over Z_TPC -- input to the corrections.

Pion+: 0-5% Centrality



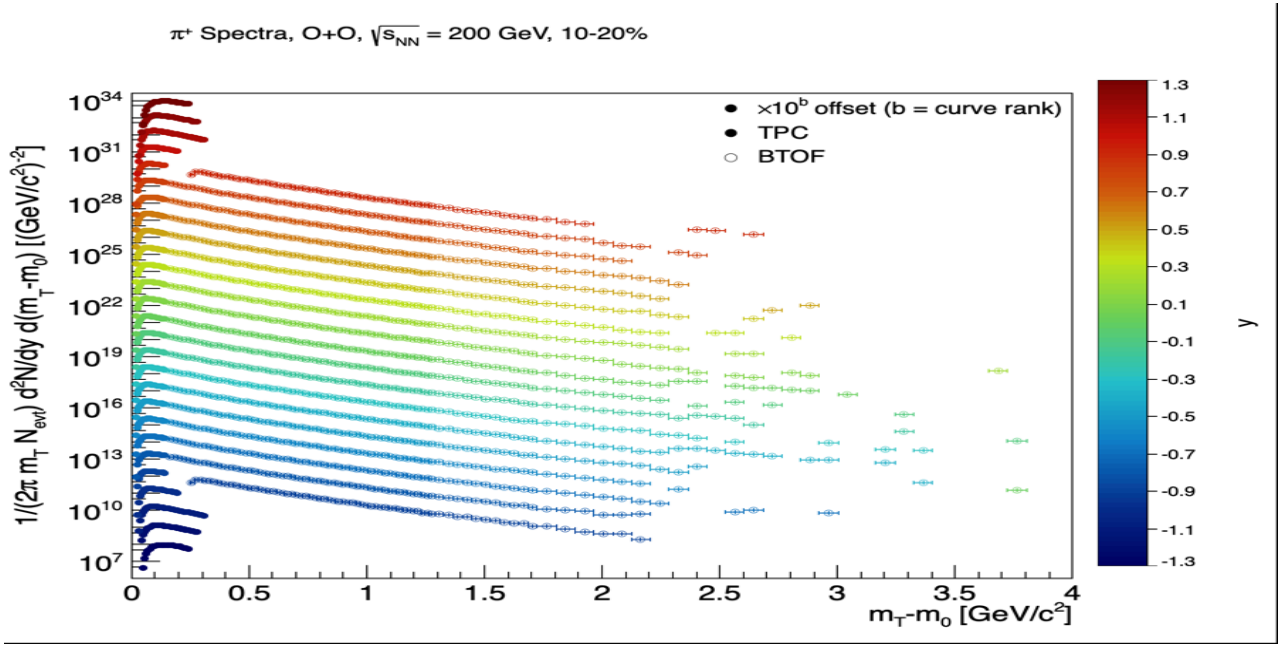
BTOF-corrected pion+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Pion+: 5-10% Centrality



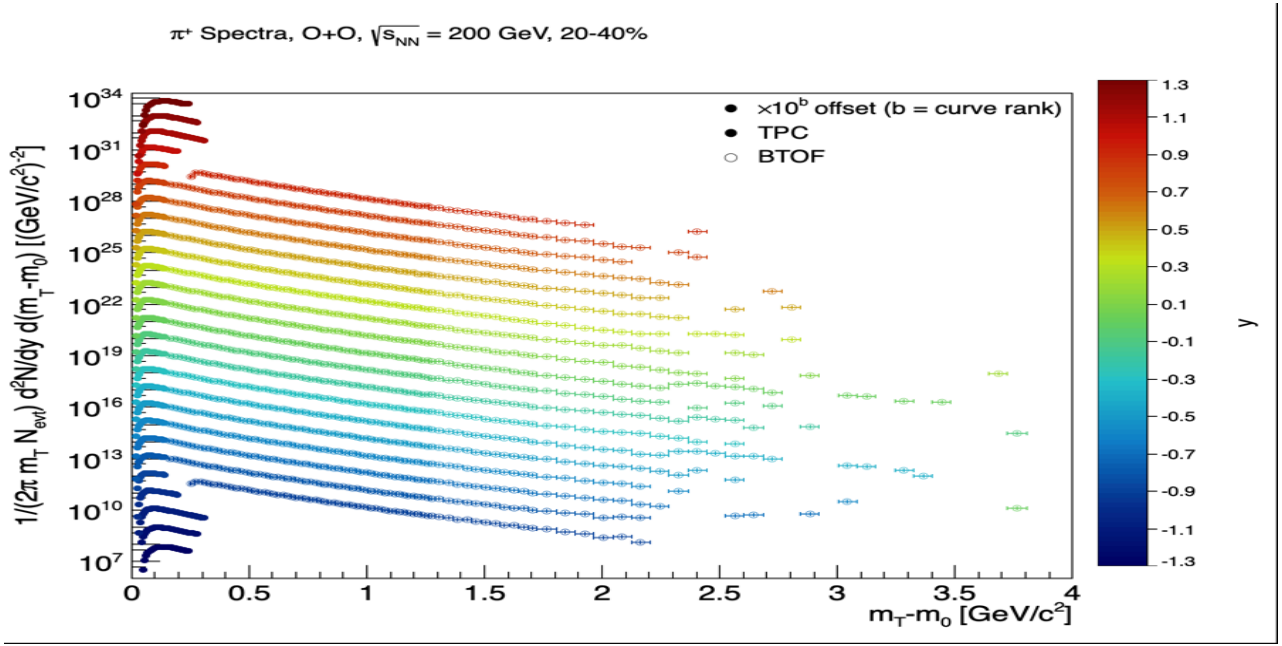
BTOF-corrected pion+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Pion+: 10-20% Centrality



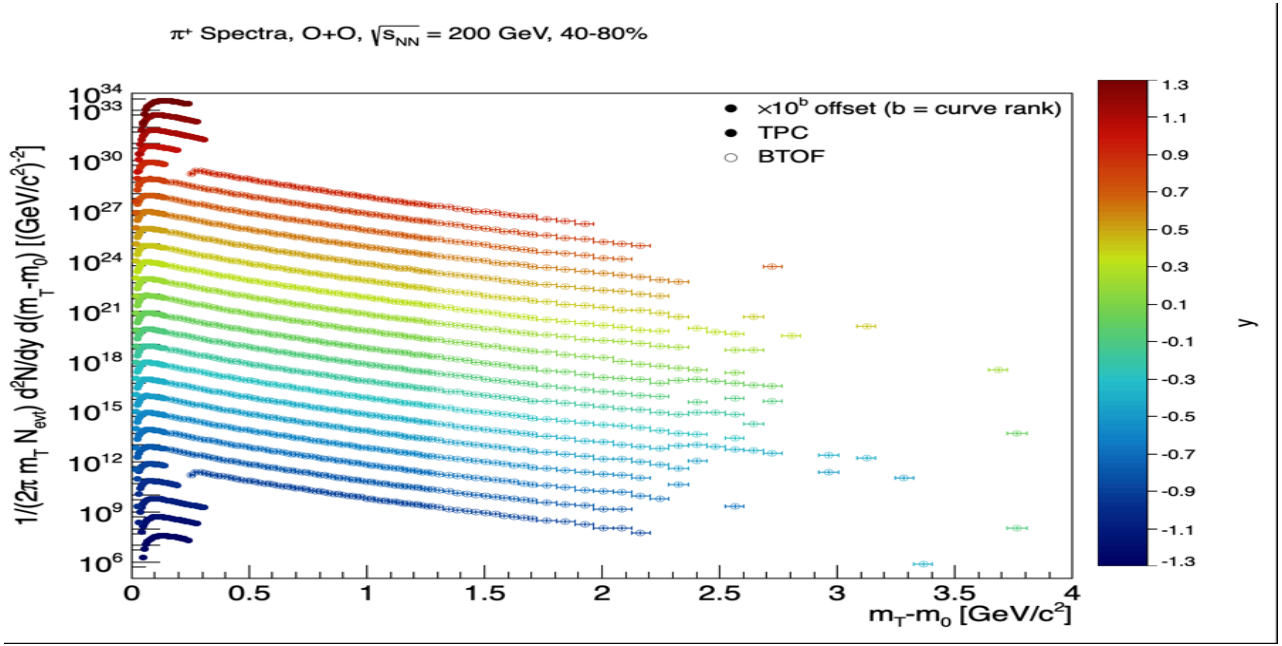
BTOF-corrected pion+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Pion+: 20-40% Centrality



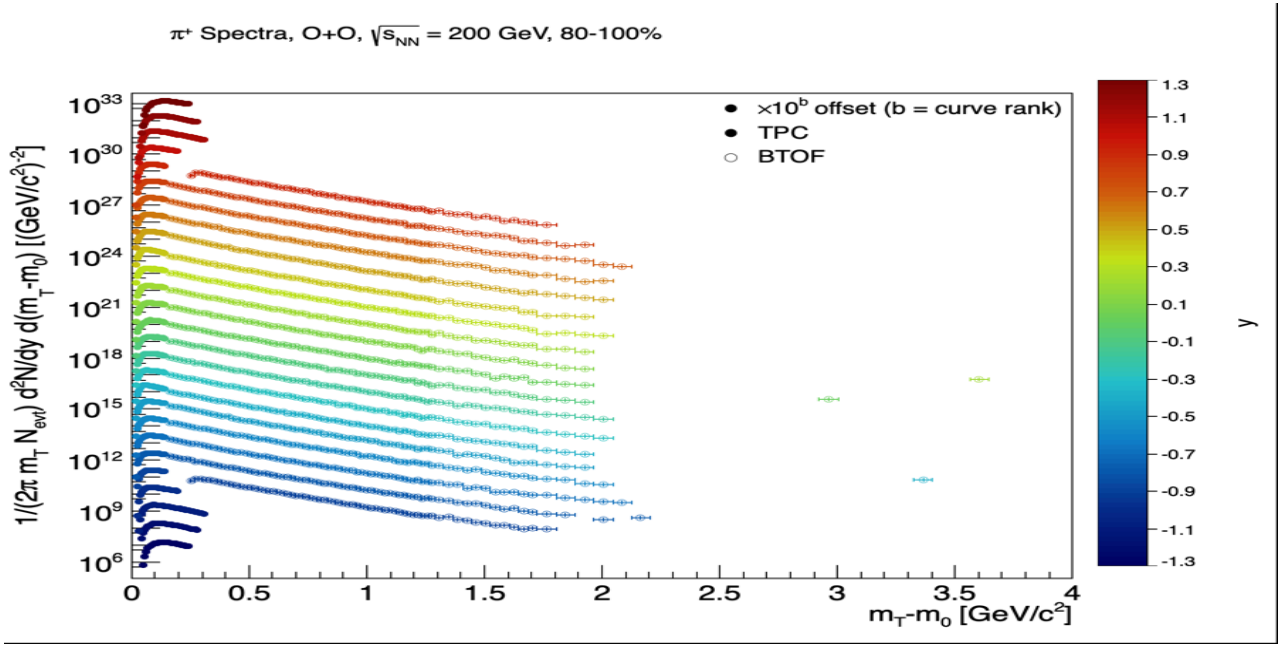
BTOF-corrected pion+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Pion+: 40-80% Centrality



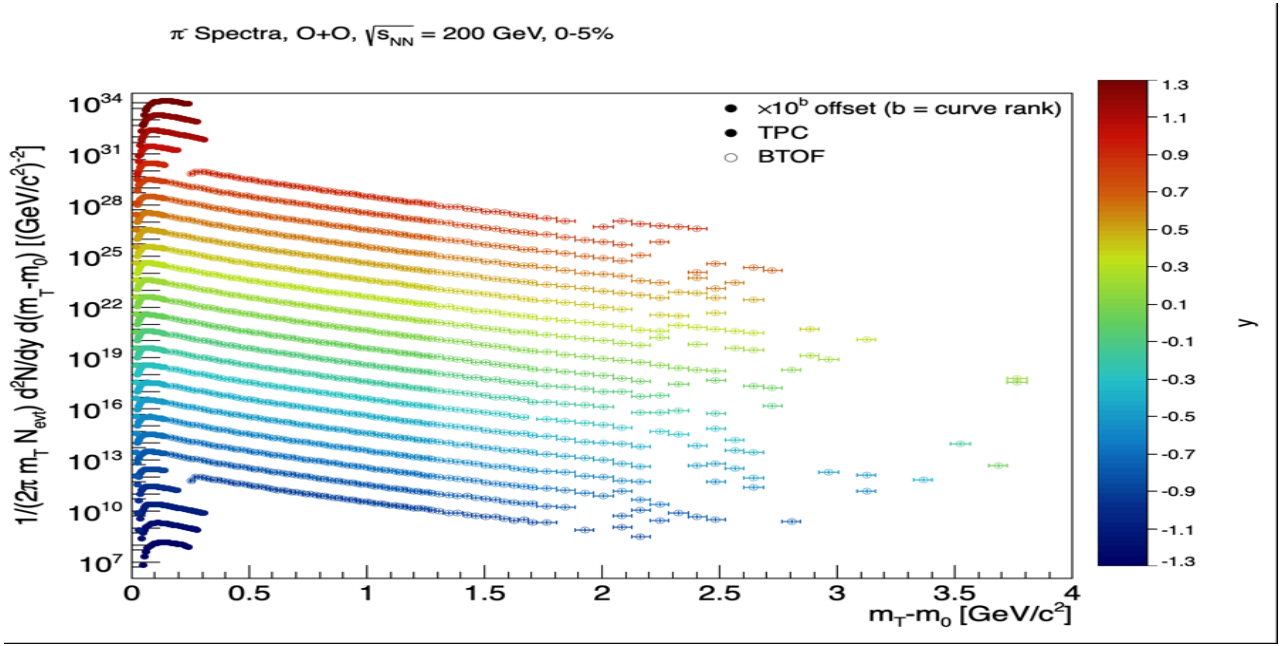
BTOF-corrected pion+ mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Pion+: 80-100% Centrality



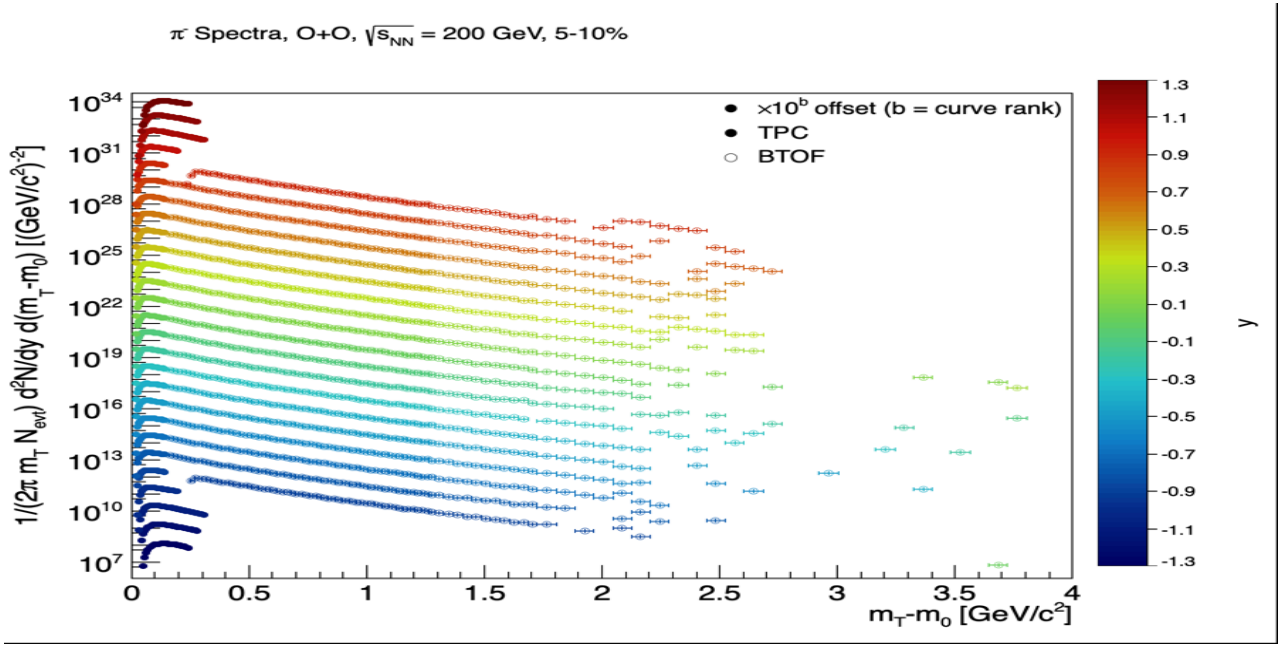
BTOF-corrected pion+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Pion-: 0-5% Centrality



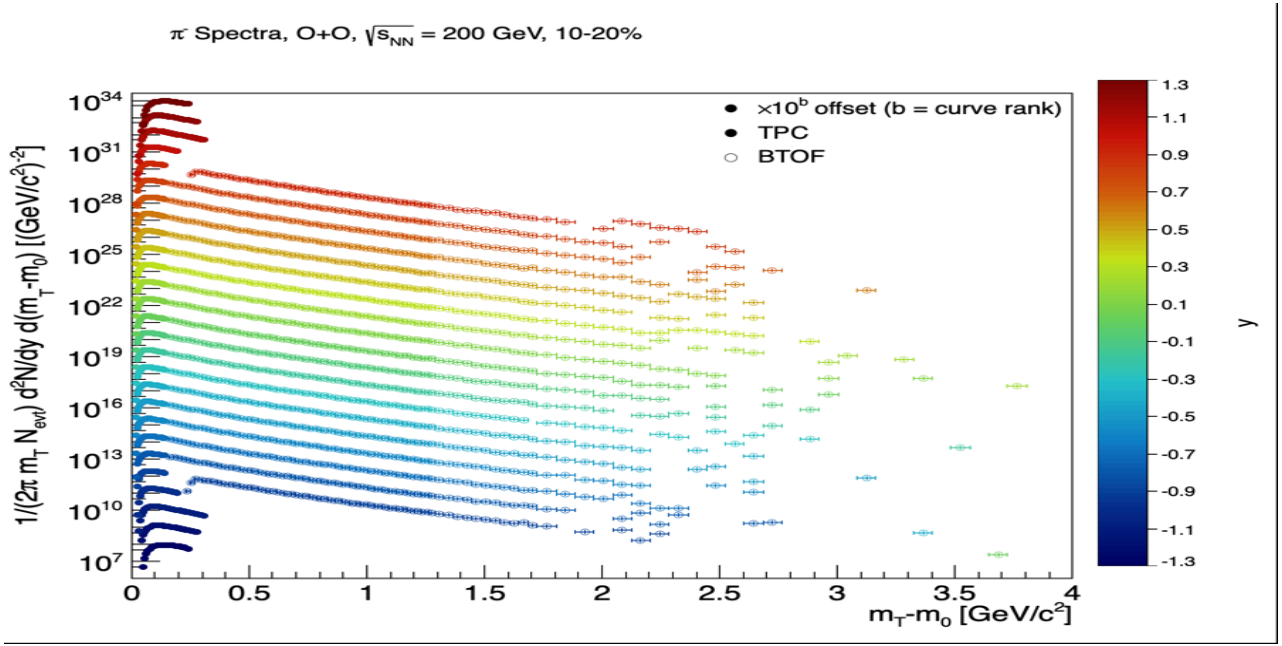
BTOF-corrected pion- mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Pion-: 5-10% Centrality



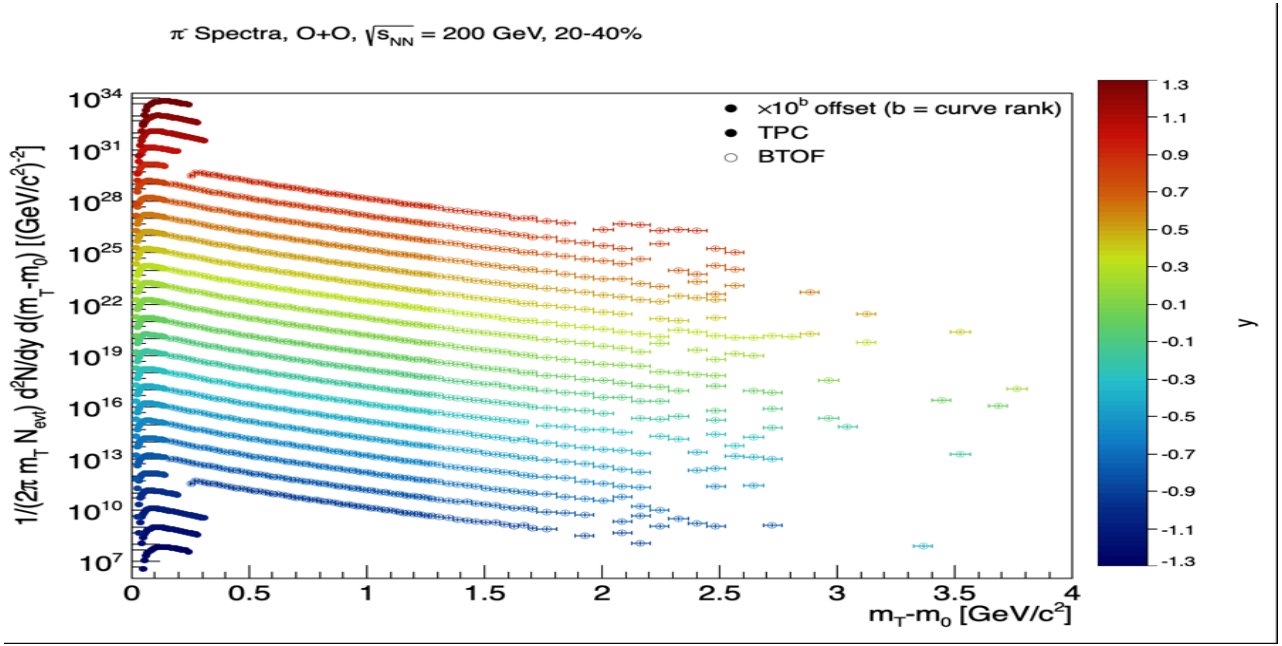
BTOF-corrected pion- mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Pion-: 10-20% Centrality



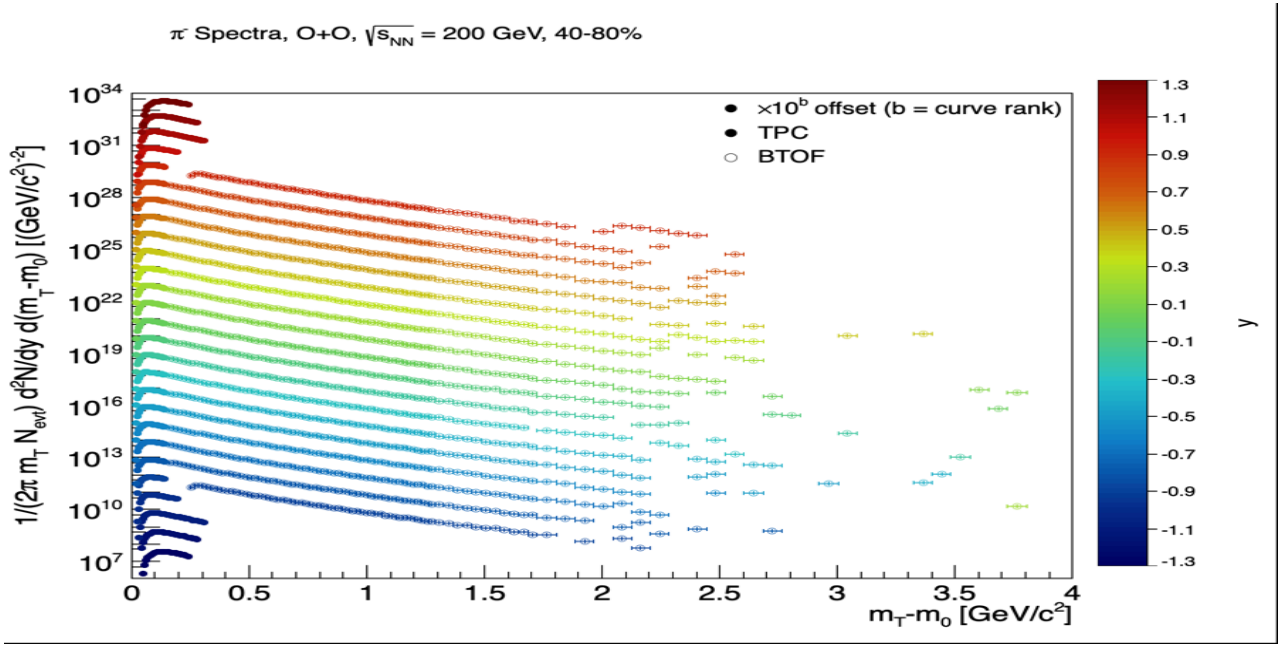
BTOF-corrected pion- mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Pion-: 20-40% Centrality



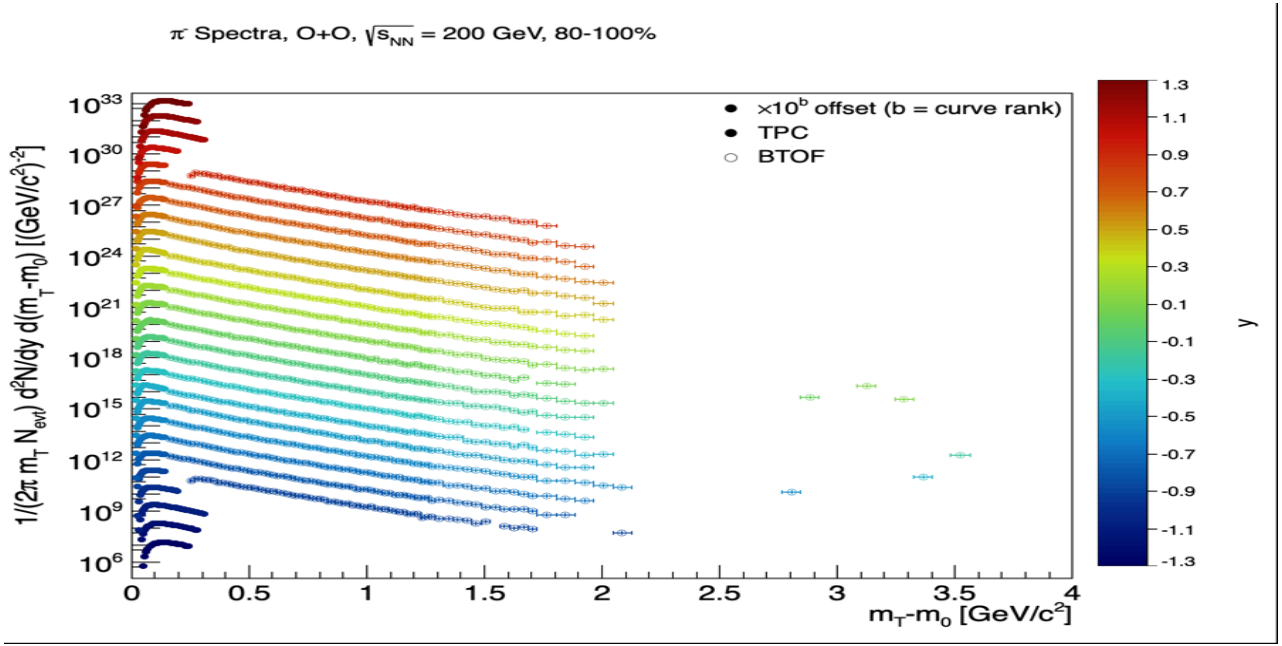
BTOF-corrected pion- mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Pion-: 40-80% Centrality



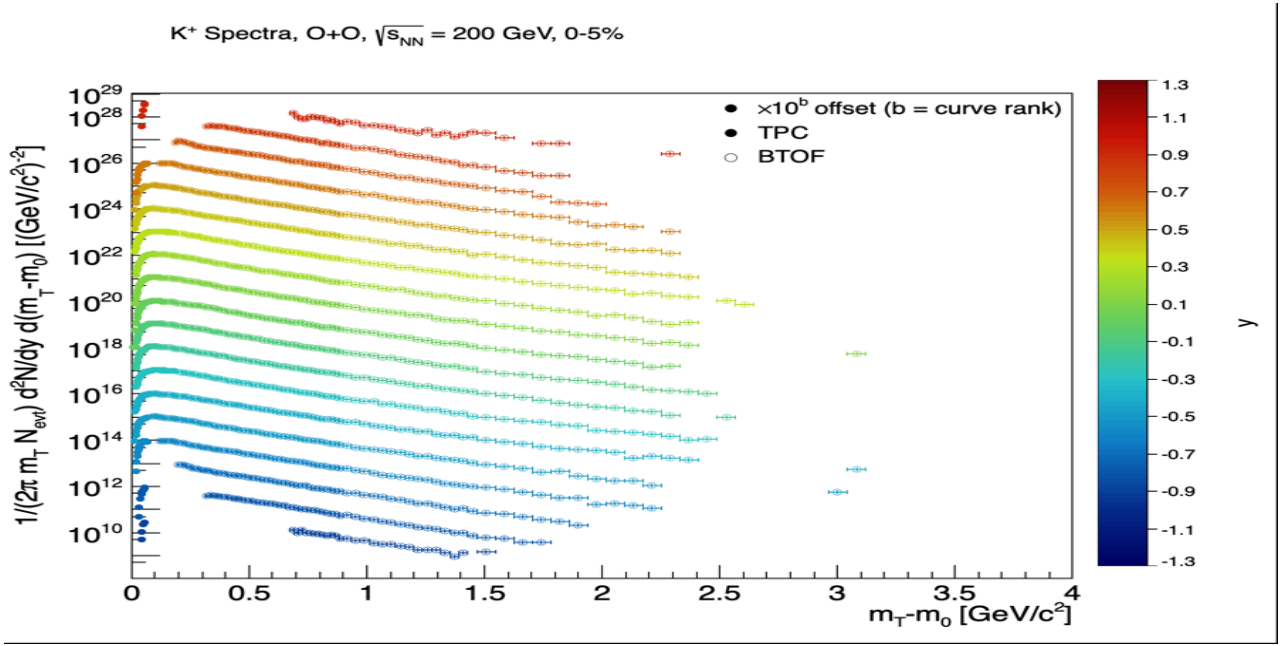
BTOF-corrected pion- mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Pion-: 80-100% Centrality



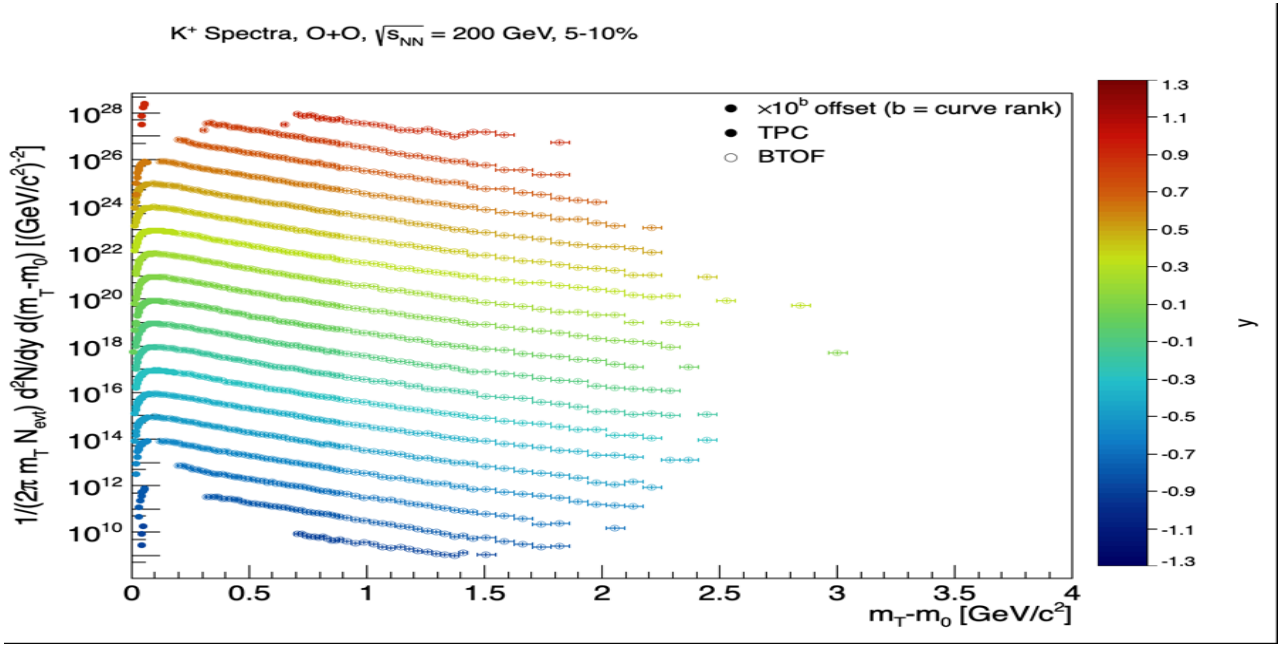
BTOF-corrected pion- mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 0-5% Centrality



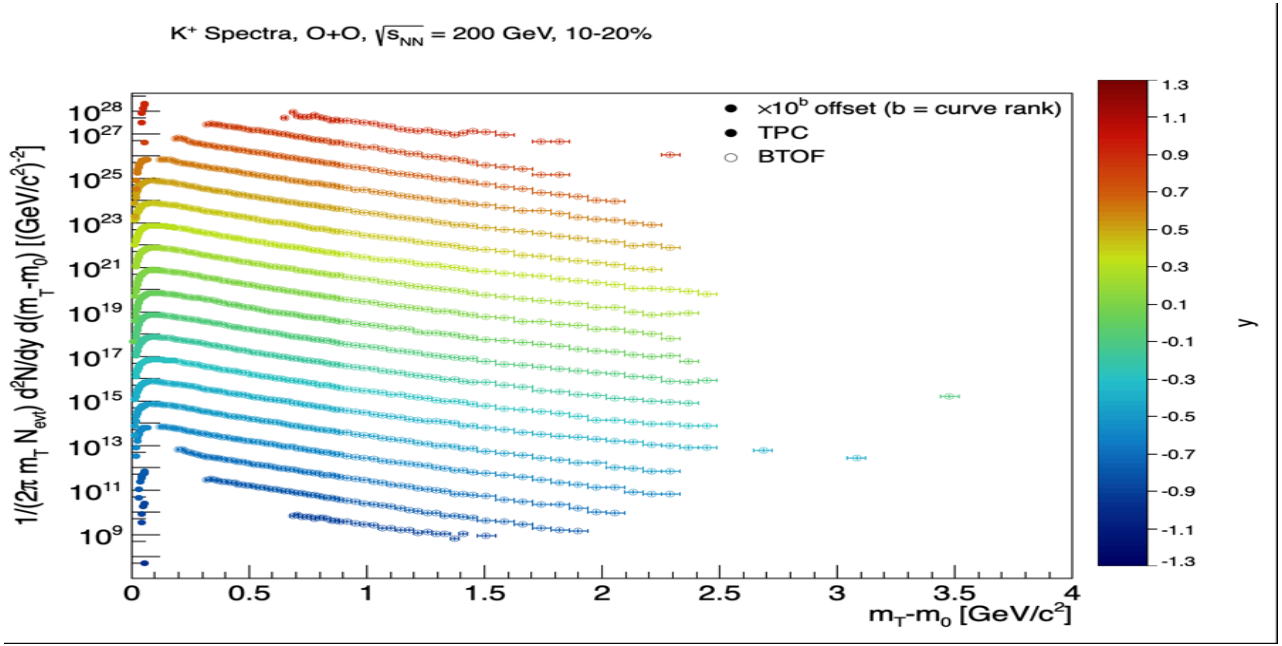
BTOF-corrected kaon+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 5-10% Centrality



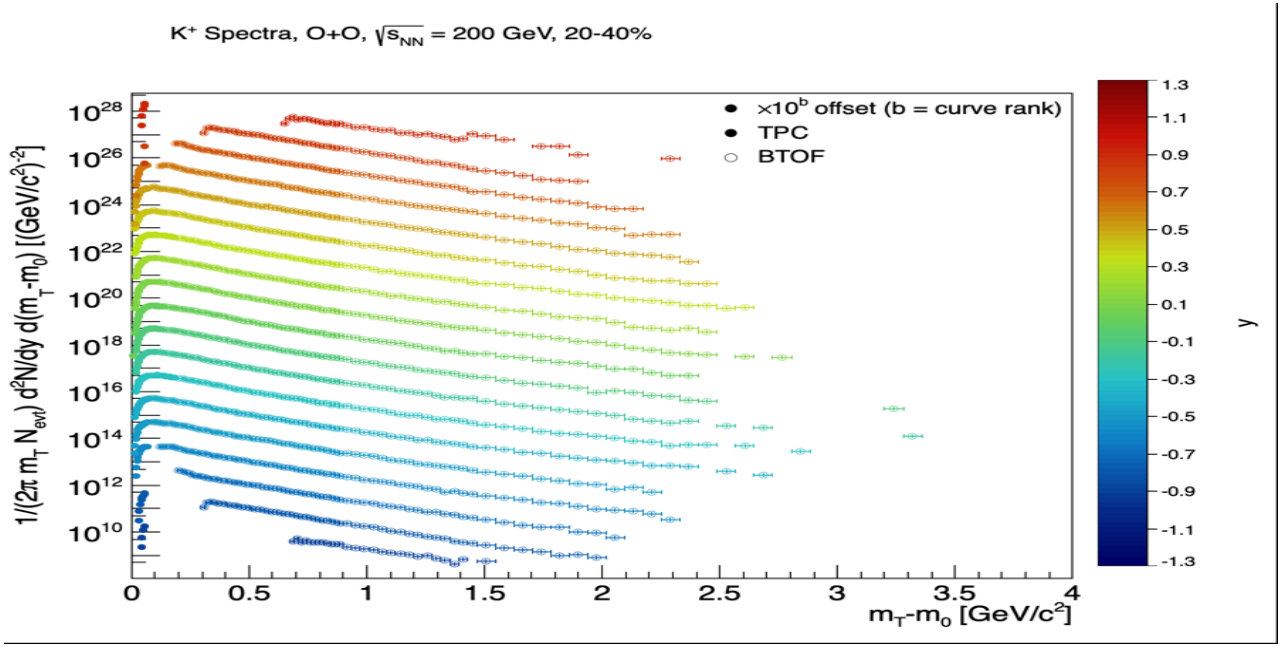
BTOF-corrected kaon+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 10-20% Centrality



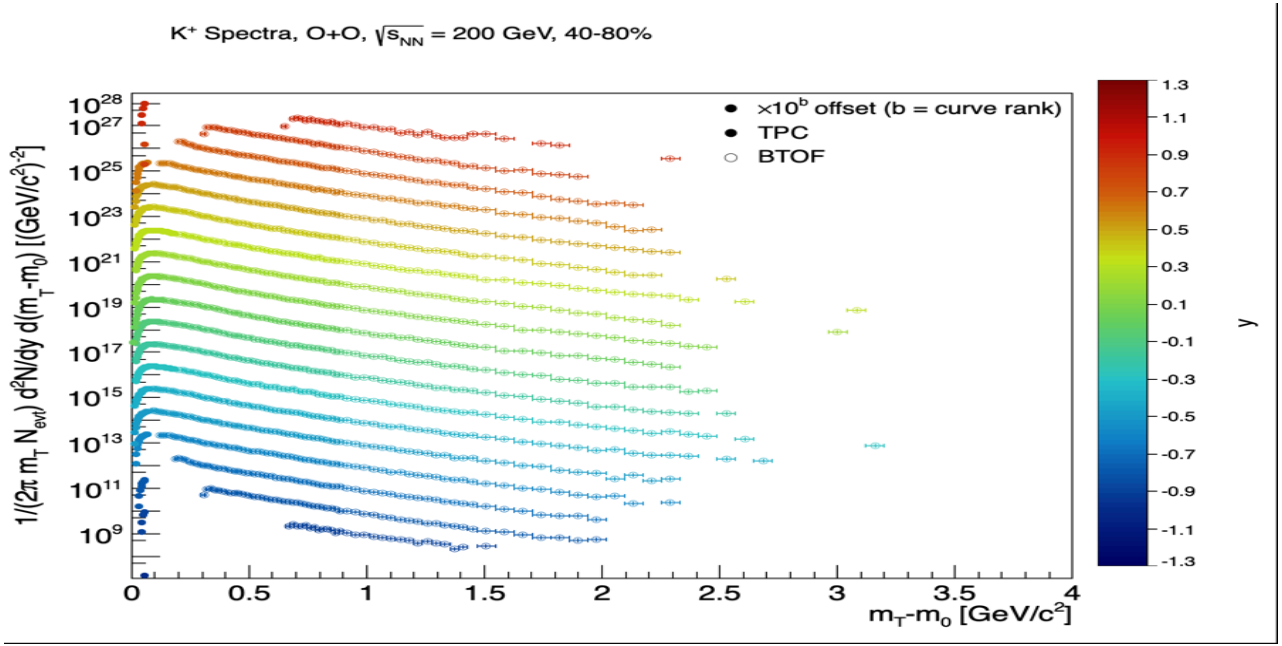
BTOF-corrected kaon+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 20-40% Centrality



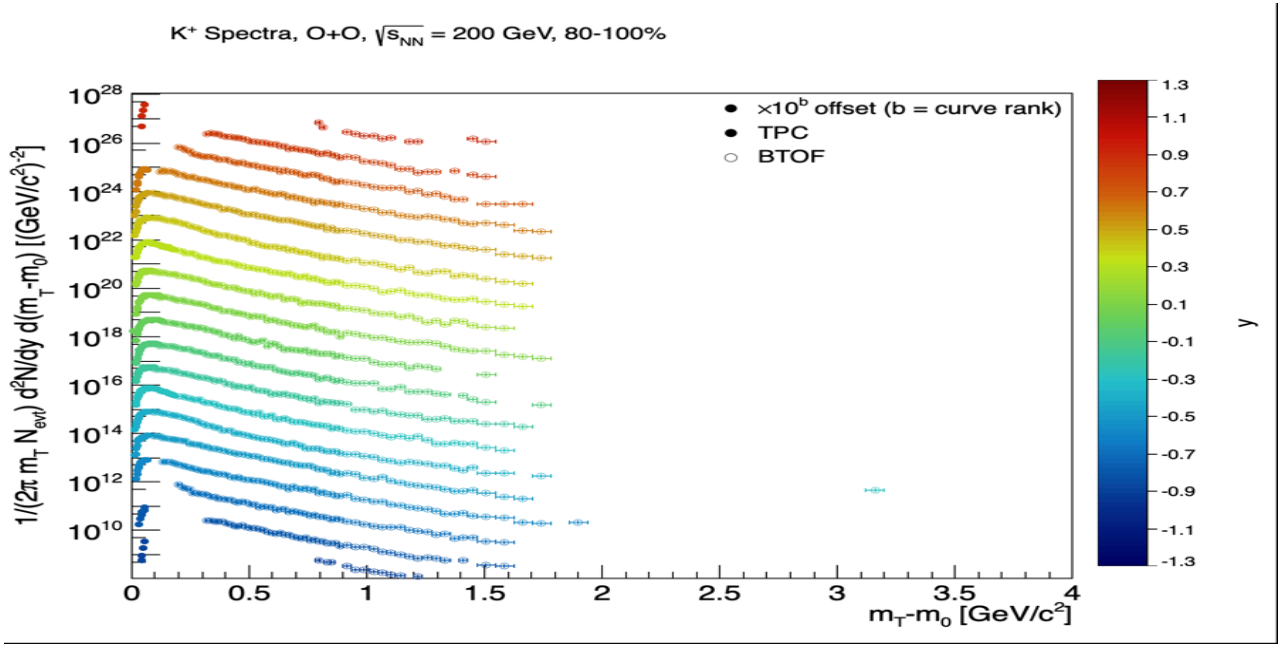
BTOF-corrected kaon+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Kaon+: 40-80% Centrality



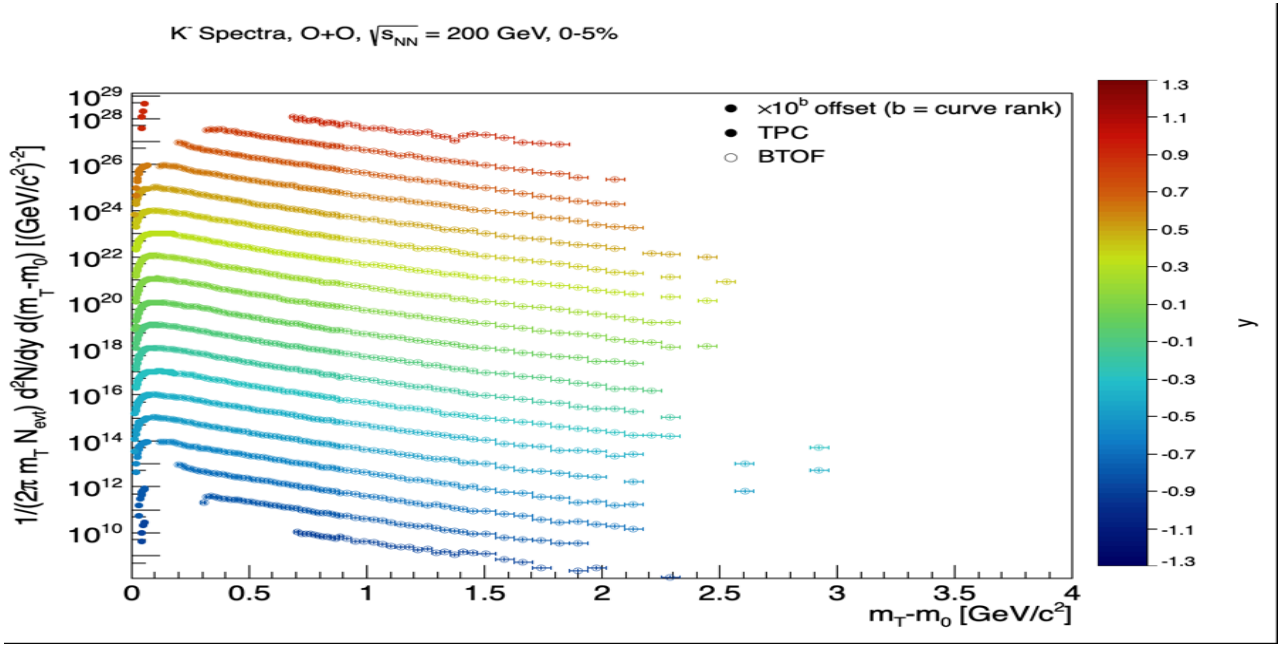
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Kaon+: 80-100% Centrality



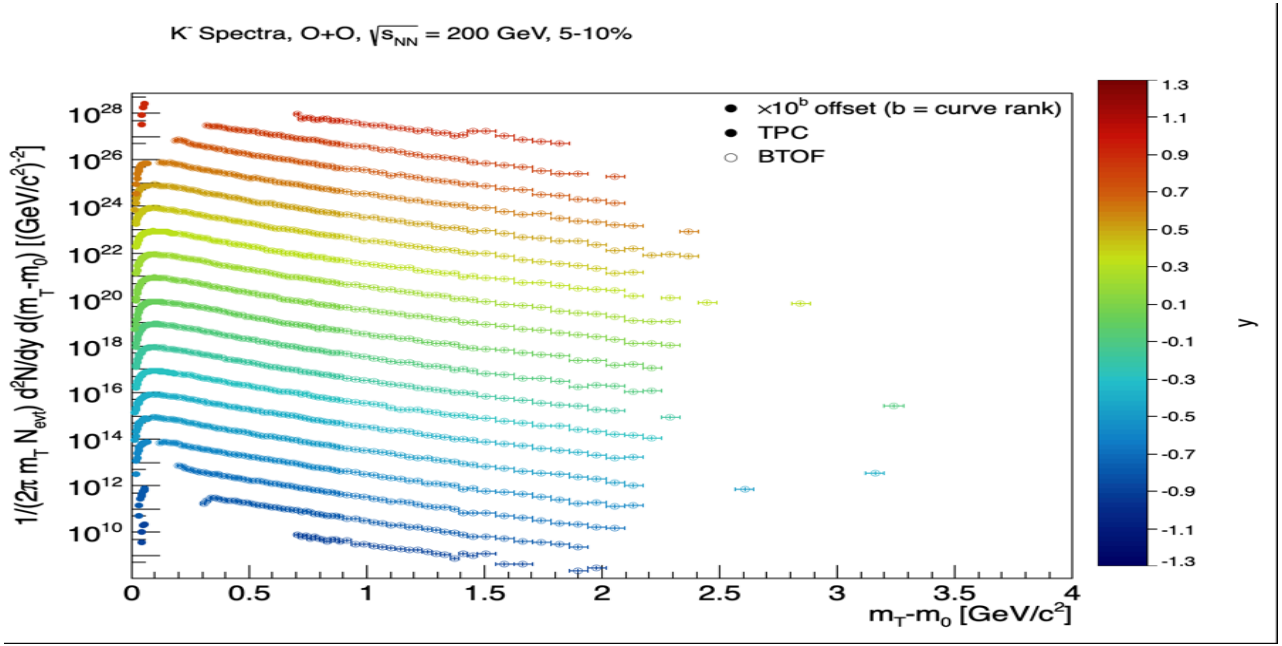
BTOF-corrected kaon+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 0-5% Centrality



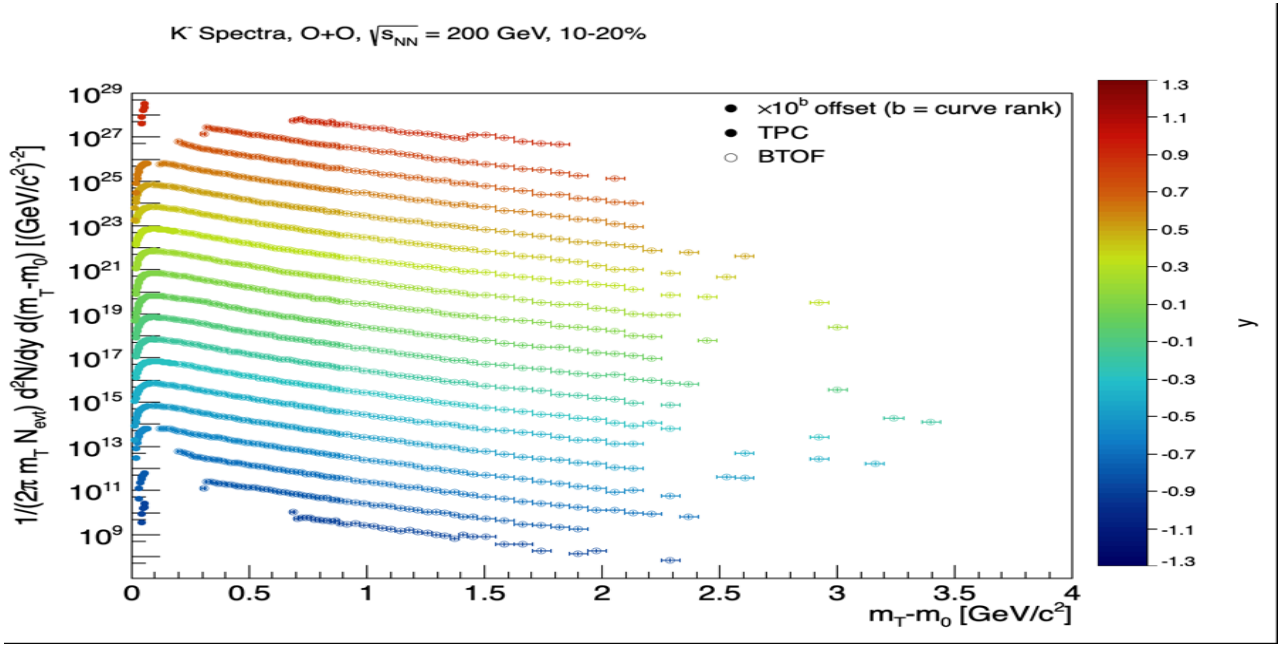
BTOF-corrected kaon- mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 5-10% Centrality



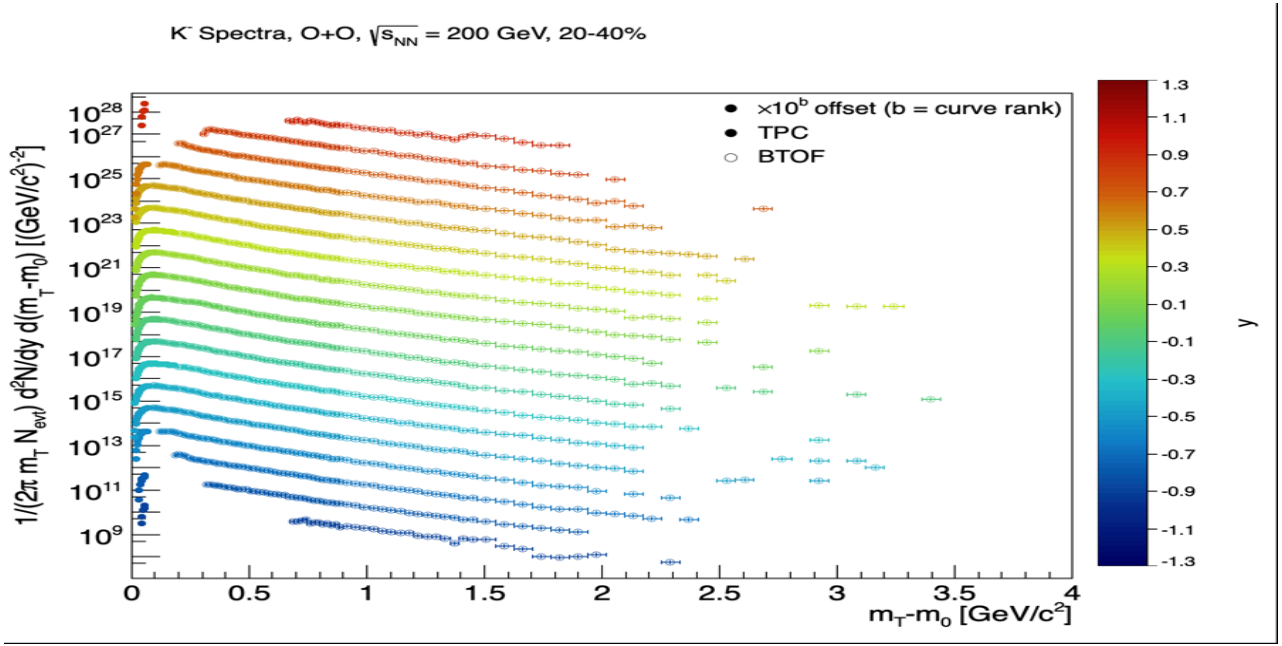
BTOF-corrected kaon- mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 10-20% Centrality



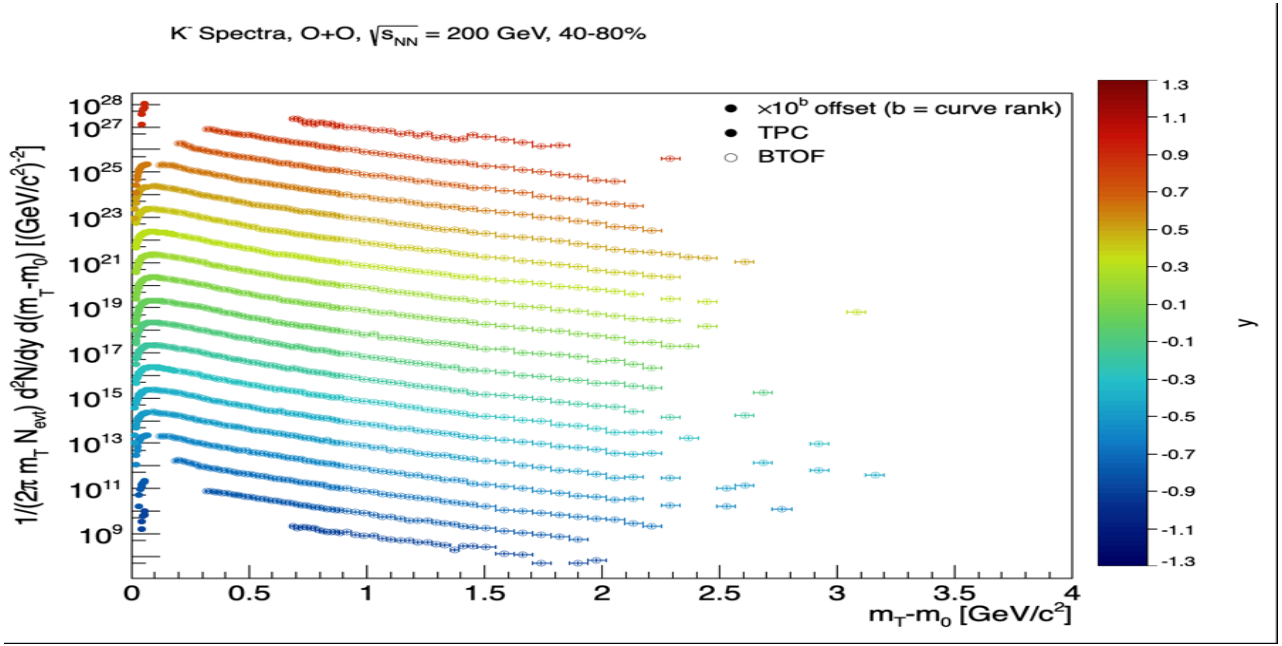
BTOF-corrected kaon- mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 20-40% Centrality



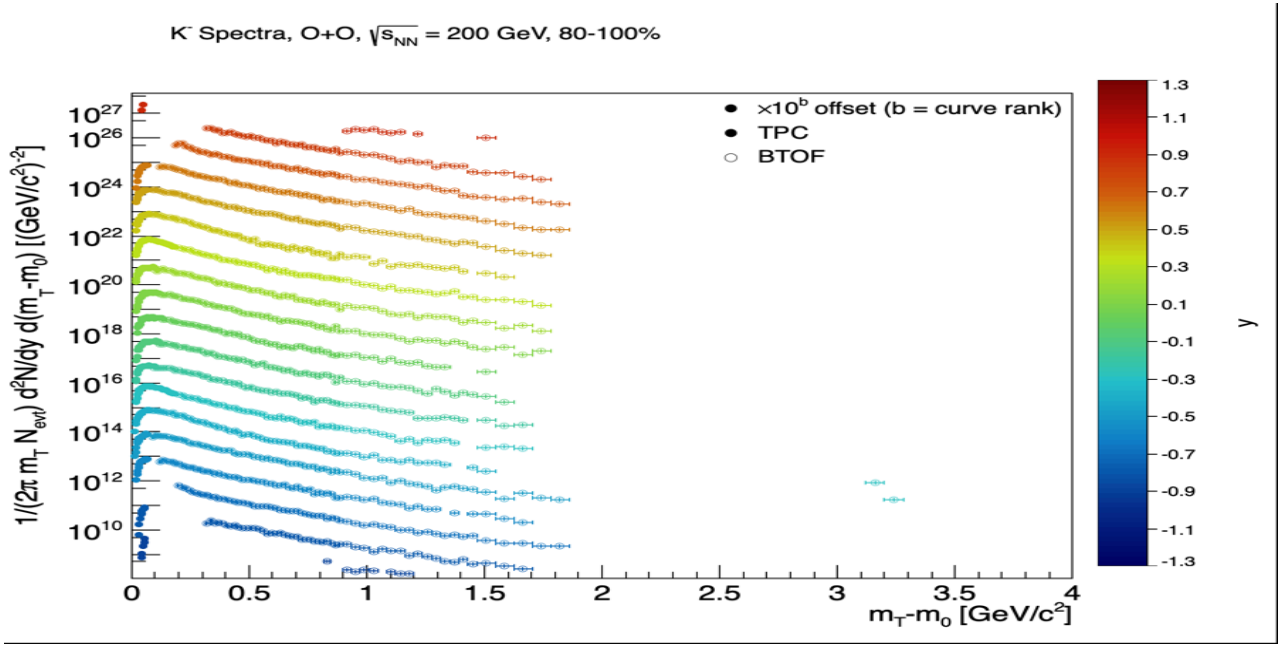
BTOF-corrected kaon- mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 40-80% Centrality



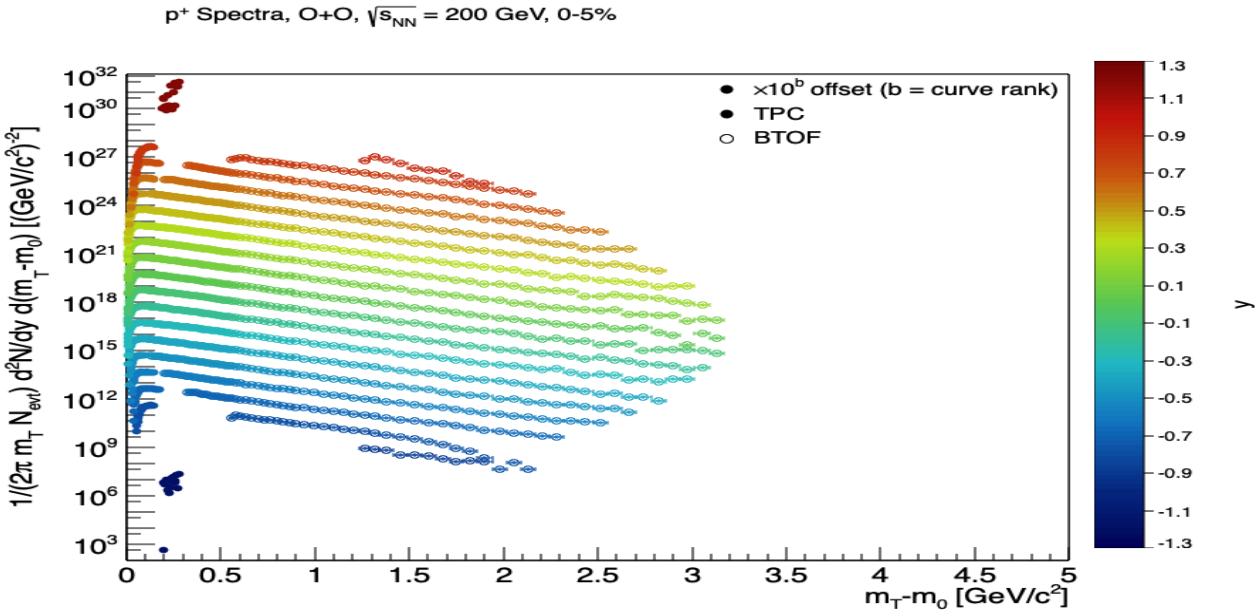
BTOF-corrected kaon- mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Kaon-: 80-100% Centrality



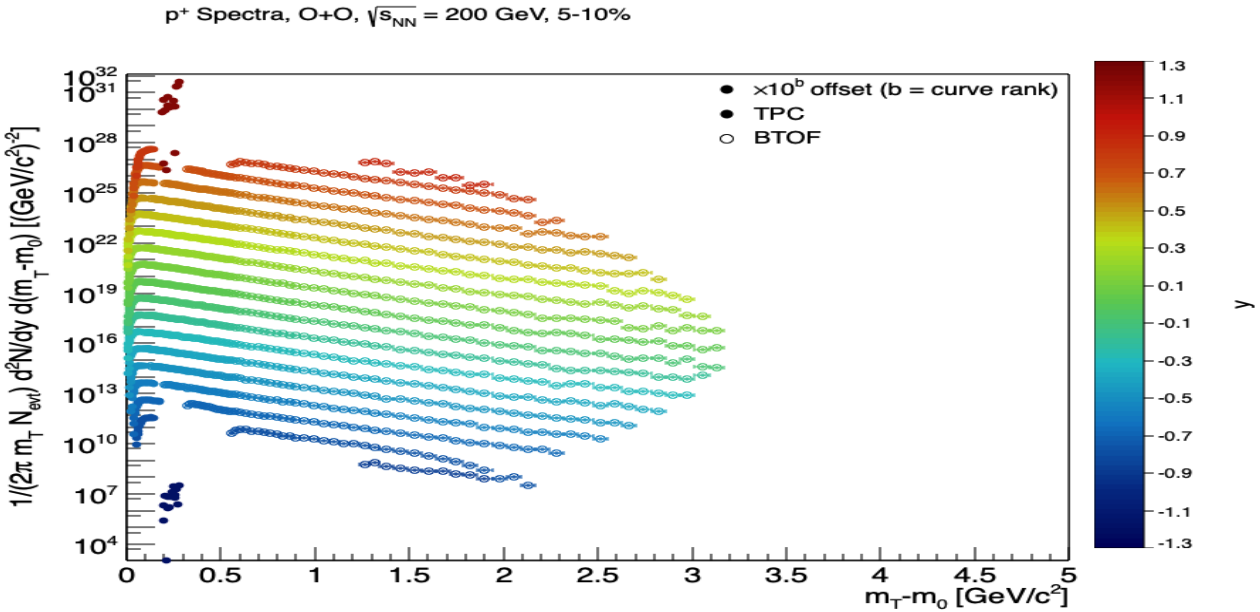
BTOF-corrected kaon- mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Proton+: 0-5% Centrality



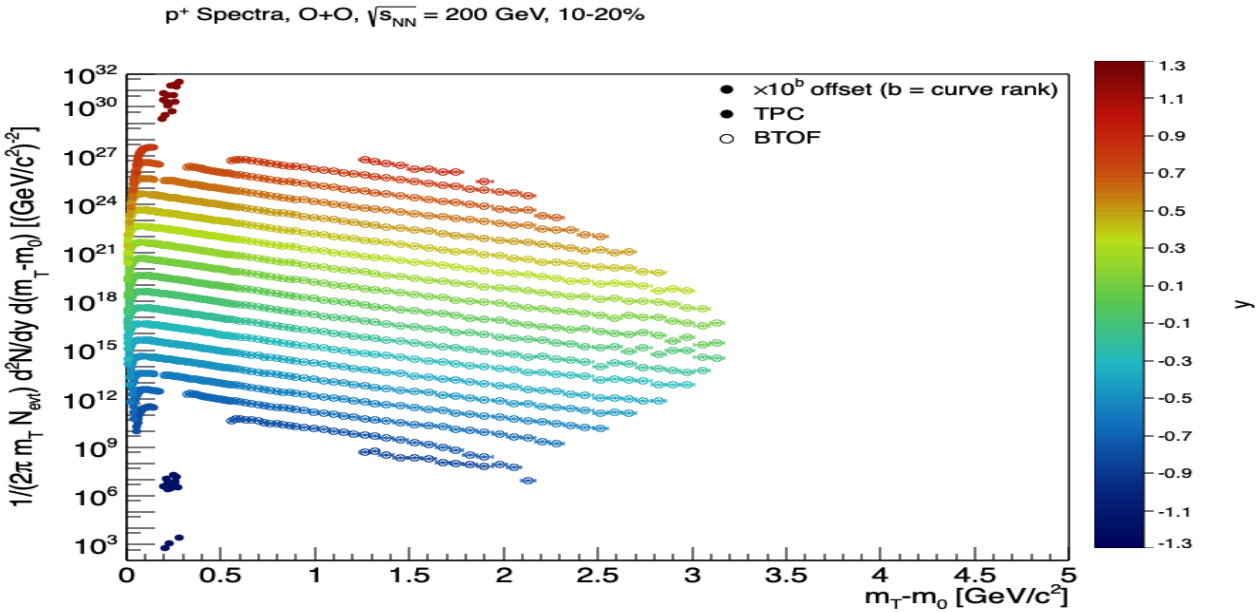
BTOF-corrected proton+ mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Proton+: 5-10% Centrality



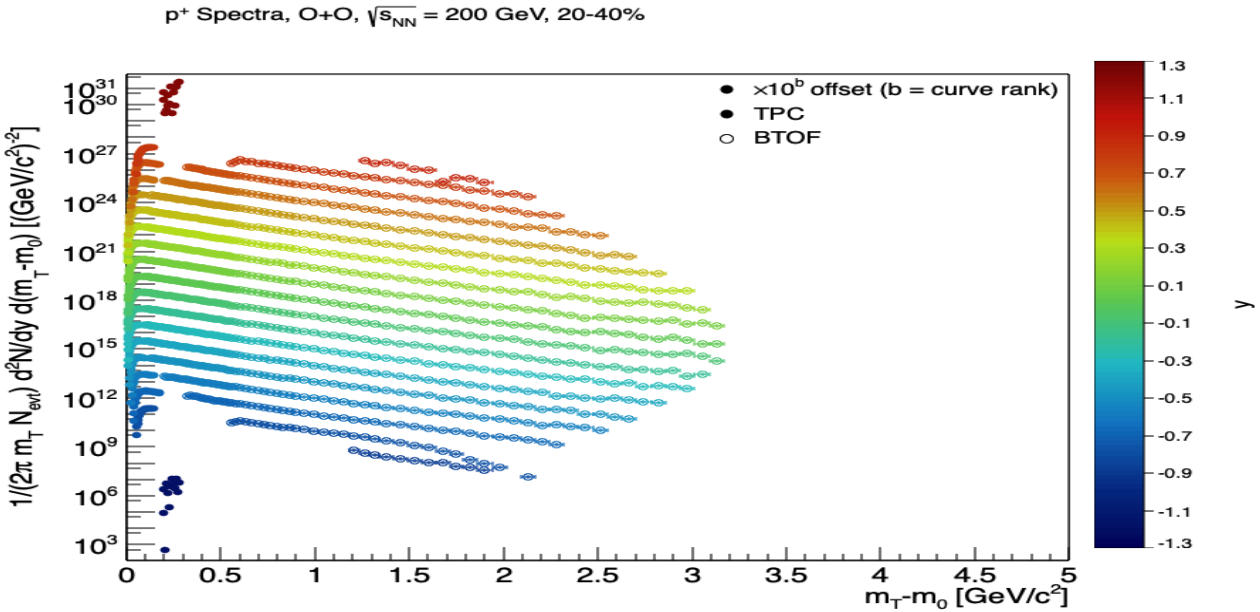
BTOF-corrected proton+ mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Proton+: 10-20% Centrality



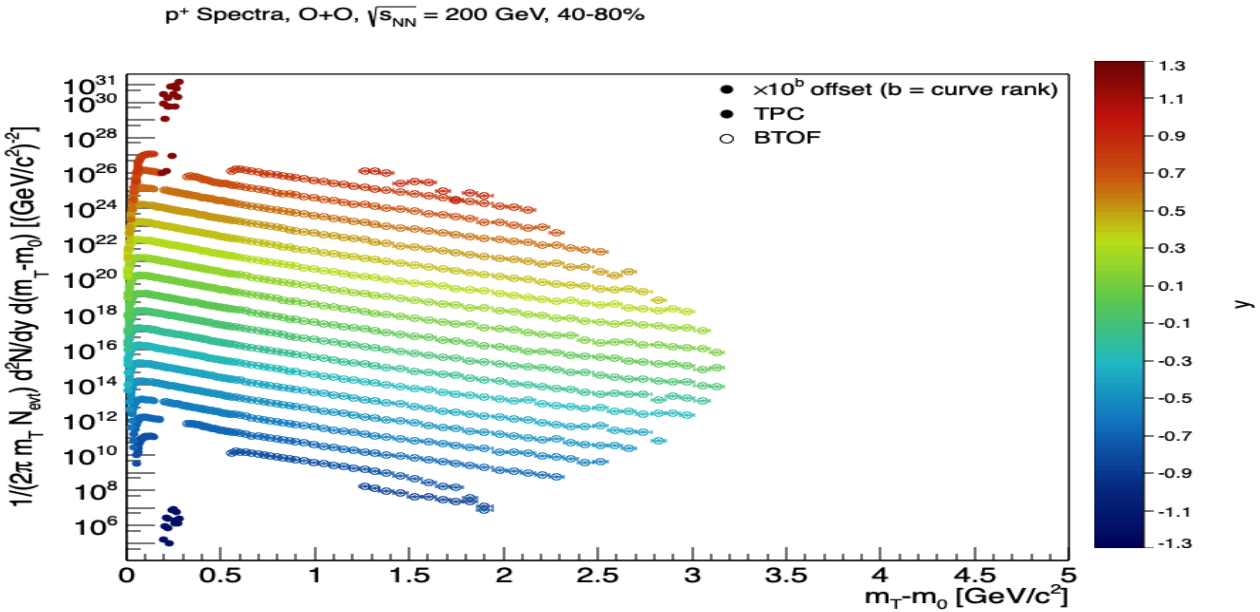
BTOF-corrected proton+ mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Proton+: 20-40% Centrality



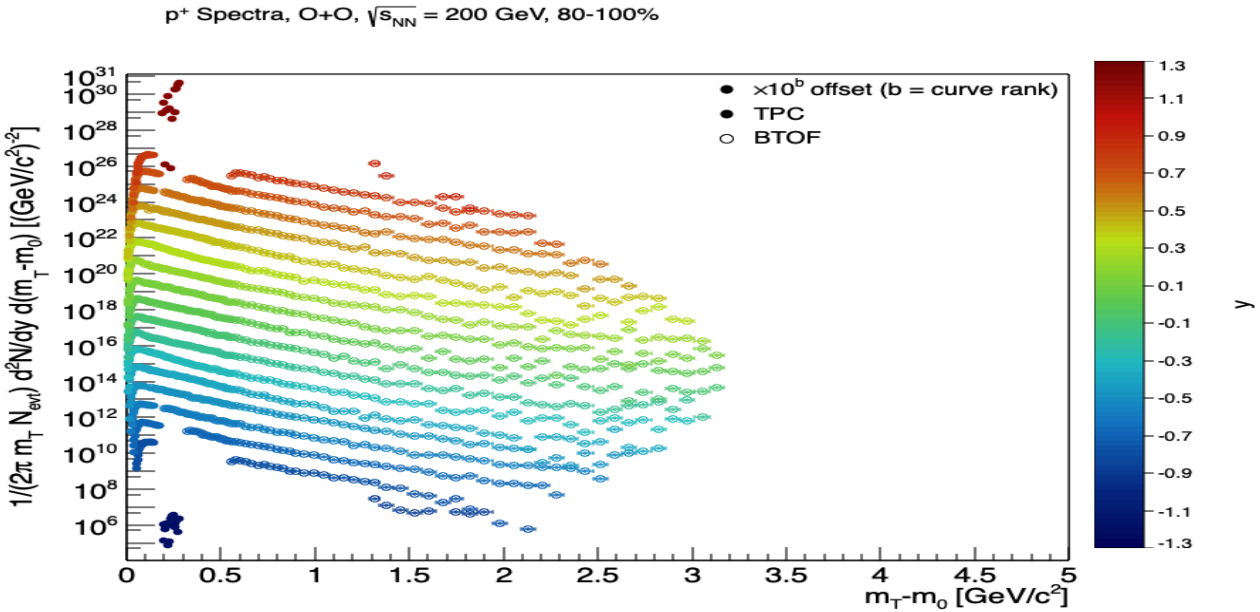
BTOF-corrected proton+ mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Proton+: 40-80% Centrality



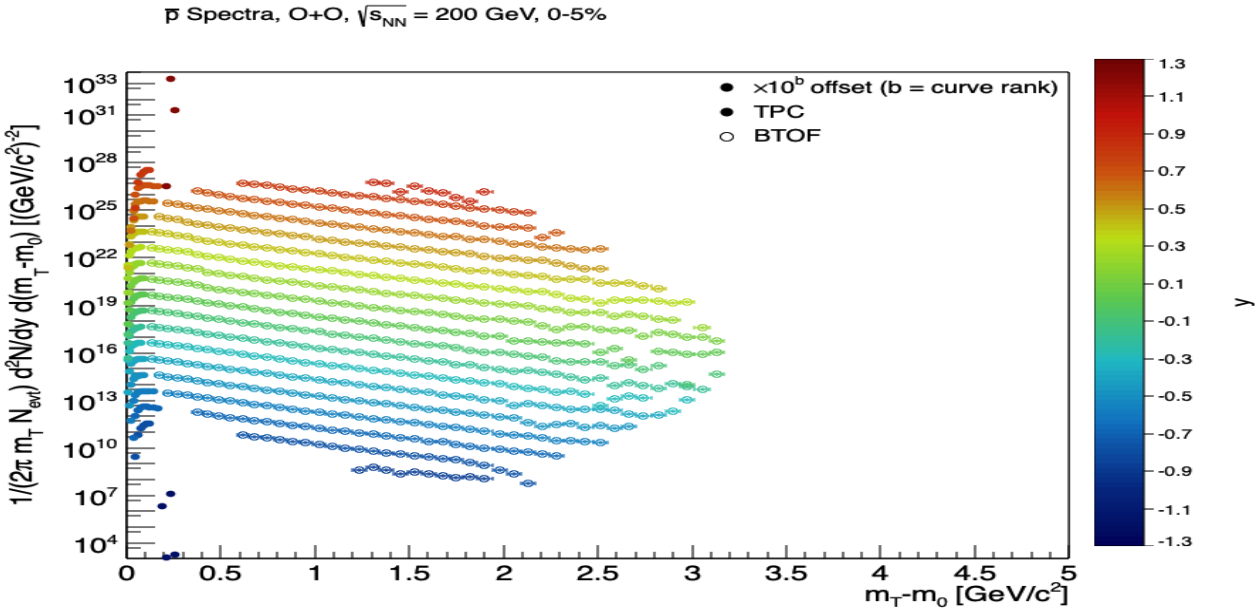
BTOF-corrected proton+ mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Proton+: 80-100% Centrality



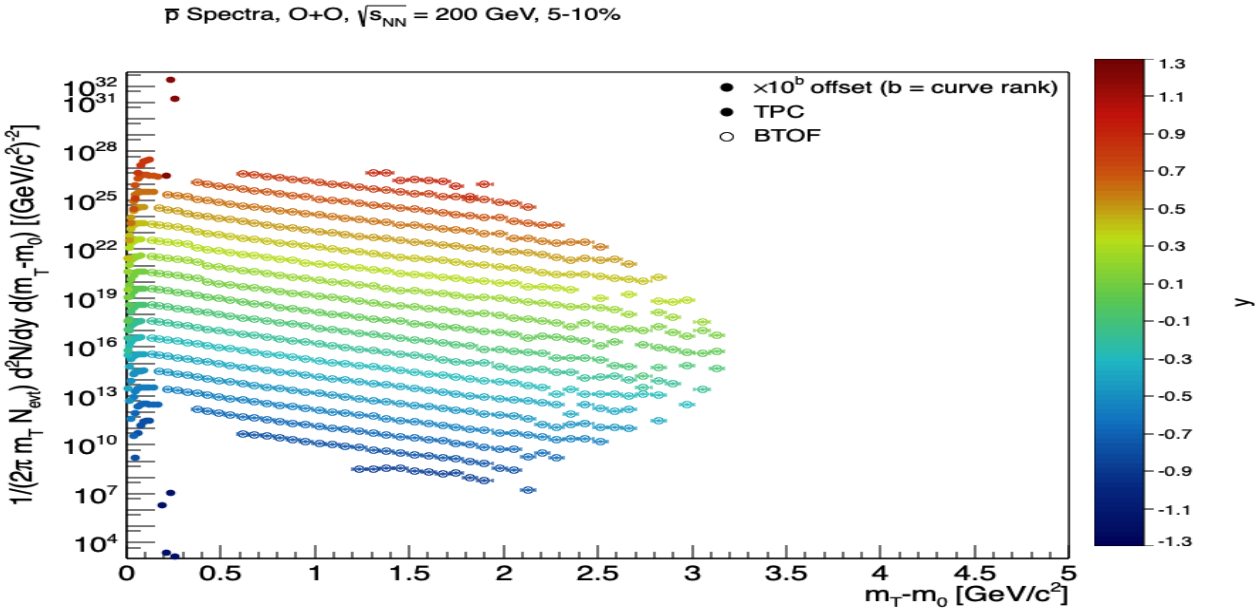
BTOF-corrected proton+ mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 0-5% Centrality



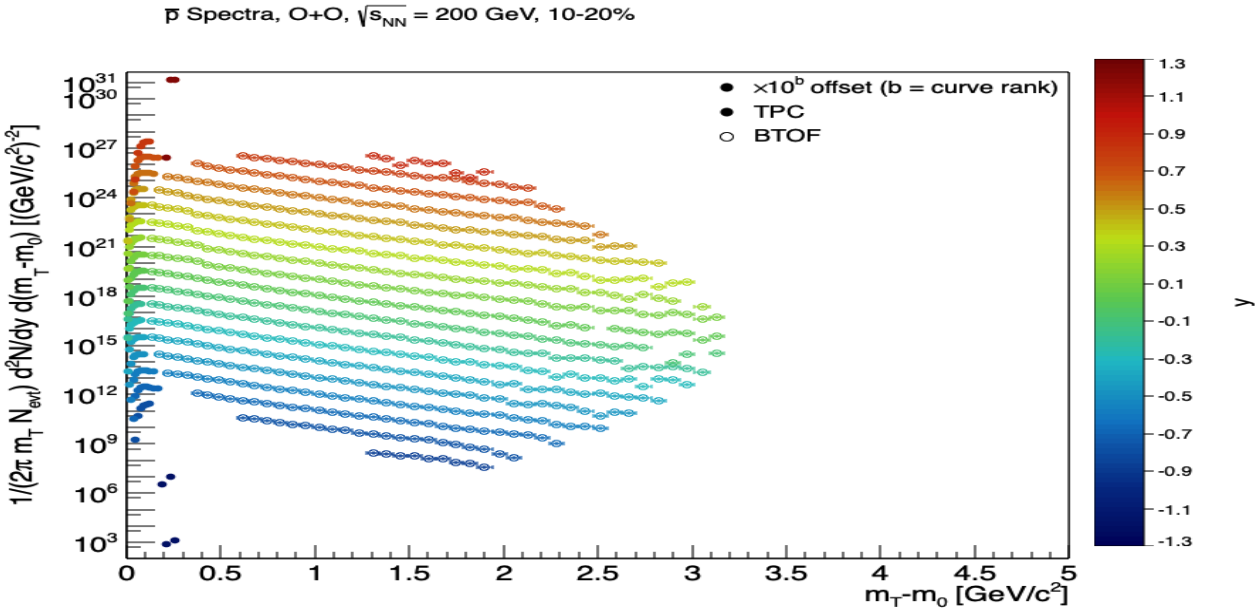
BTOF-corrected antiproton mT-m0 spectra, 0-5% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 5-10% Centrality



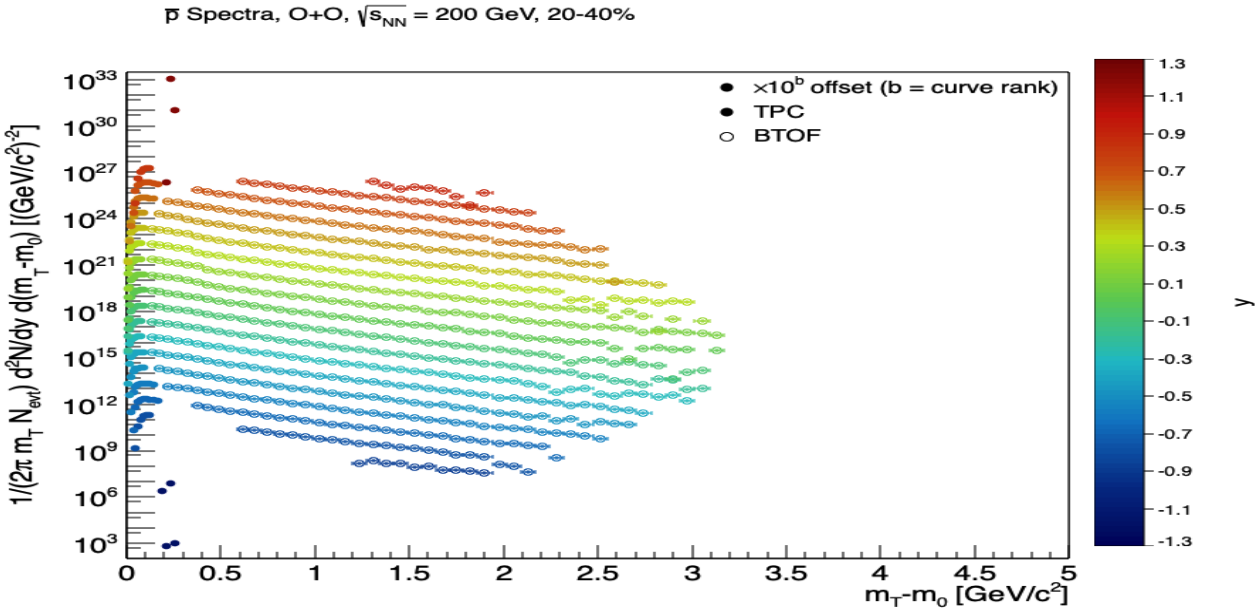
BTOF-corrected antiproton mT-m0 spectra, 5-10% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 10-20% Centrality



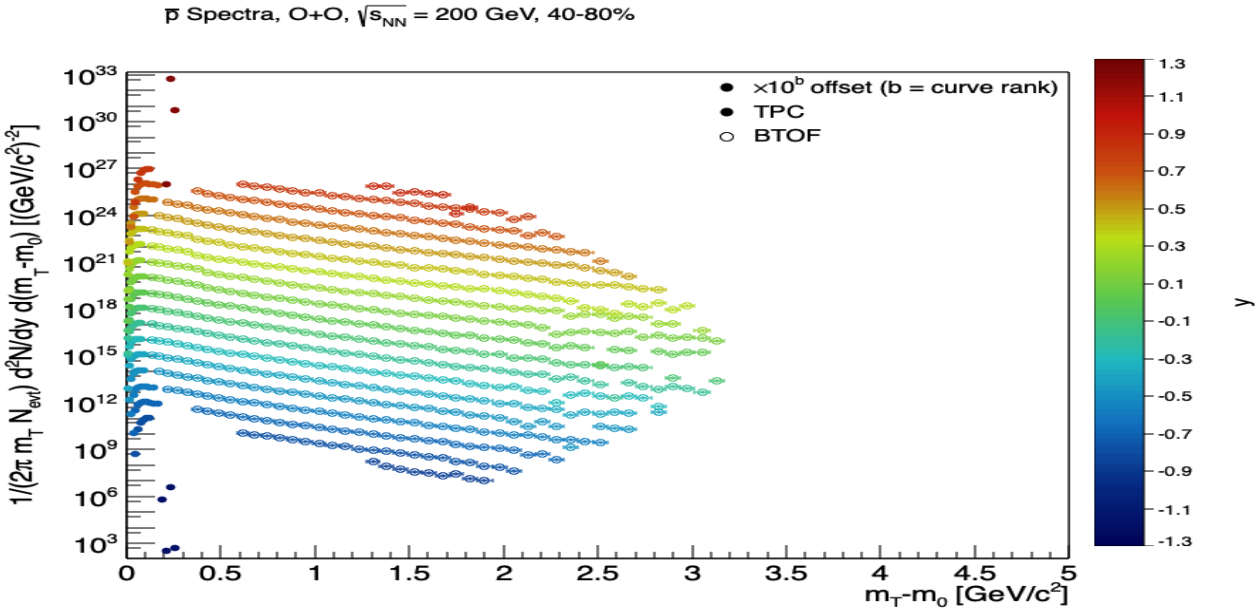
BTOF-corrected antiproton mT-m0 spectra, 10-20% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 20-40% Centrality



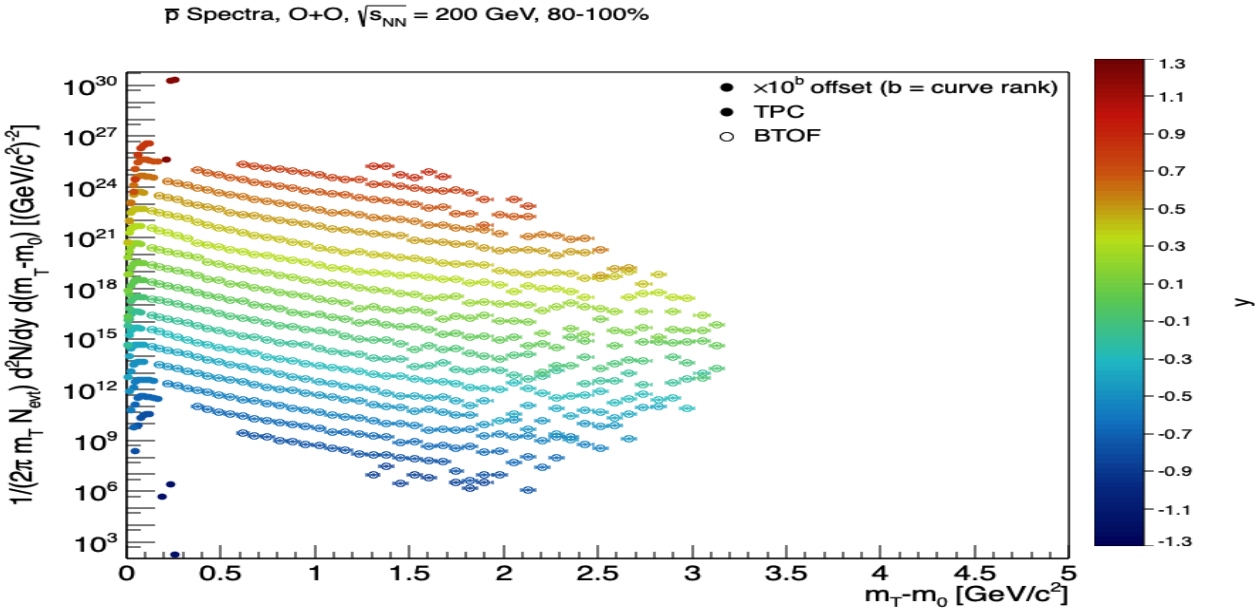
BTOF-corrected antiproton mT-m0 spectra, 20-40% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 40-80% Centrality



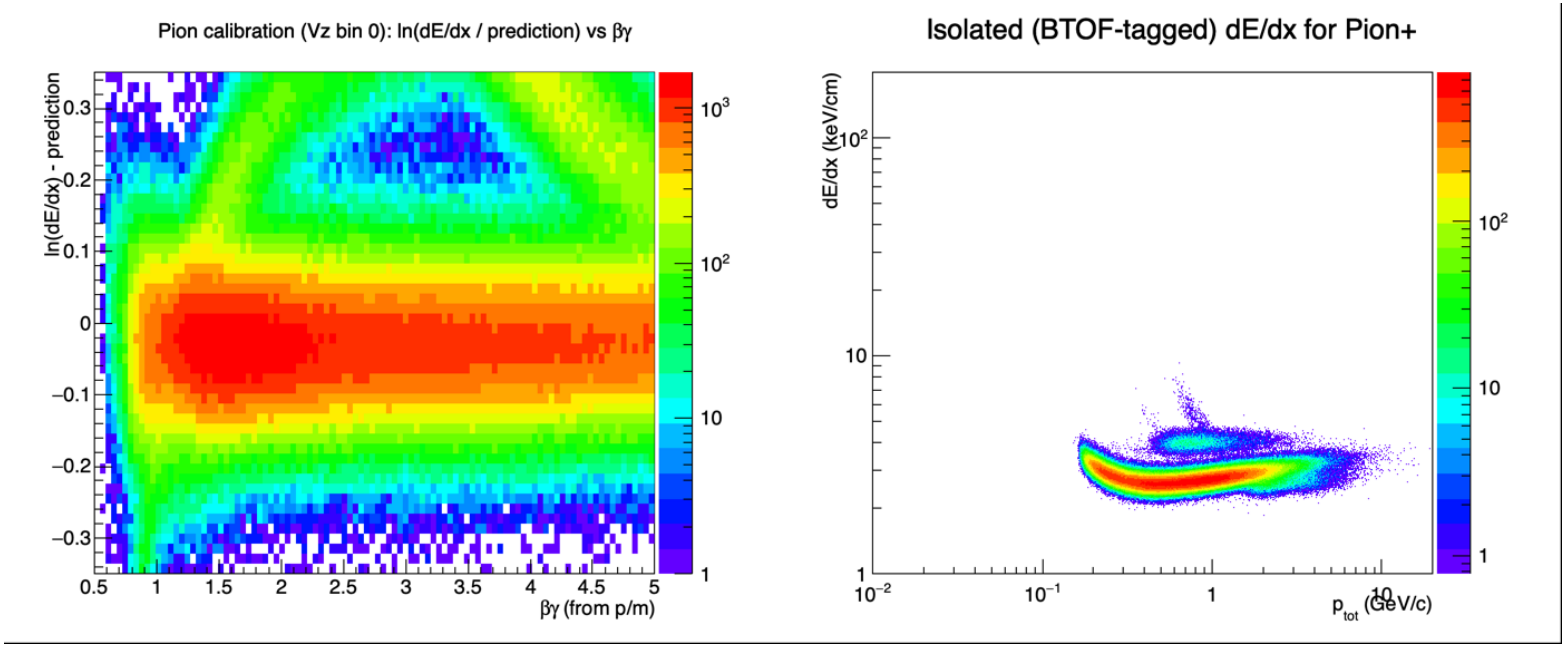
BTOF-corrected antiproton mT-m0 spectra, 40-80% centrality, TPC+BTOF, all rapidity bins.

Antiproton: 80-100% Centrality



BTOF-corrected antiproton mT-m0 spectra, 80-100% centrality, TPC+BTOF, all rapidity bins.

dE/dx Calibration QA



Pion dE/dx calibration residual vs beta-gamma (left) and isolated BTOF-tagged pion+ dE/dx (right) -- calibration is flat and centered.